Q-COMEM: QUANTIZED SPLIT REPLAY FOR MEMORY-CONSTRAINED HYBRID LLMS

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ABSTRACT

Repeated queries over stable documents create a capacity problem for memory-constrained inference: exact prefix caching minimizes warm latency but retains state across all layers, whereas dense recomputation retains nothing but repeatedly processes the document. We present Q-COMEM, a capacity-first extension of CoMem split replay for hybrid LLMs. An offline Write runs the document to a split depth and group-quantizes the boundary residual together with the lower-layer KV, convolution, and recurrent state later queries need; online Read dequantizes and forks that state, keeps mutable request state private, and replays the query before reconstructing the suffix. A fail-closed FORKAUDIT trace validates immutability, copy-on-write, recurrent rebinding, and call provenance because output equality alone can miss state aliasing; that factorial runs full-prefix BF16 KV with no split depth and no quantization, so it audits the ownership discipline and not the quantized Read path.

On an archival 60-item Qasper/2WikiMQA validation cohort with Qwen3.5-35B-A3B and H20-3e, a frozen Q4/Q4/Q8 policy reduces mean retained tensor payload from 136.235 to 9.661 MiB/document ($14.10\times$, 60-item mean). That reference is what the unmodified stack retains, in its native dtypes; 30.000 MiB/document of it is FP32 GDN recurrent state in excess of a BF16 encoding, so part of the gain is dtype narrowing rather than packing, and against an all-BF16 reference of 106.235 MiB/document the policy is $10.9965\times$ smaller. Against the same full-prefix reference, its mean F1 (0–100) is 54.24 versus 54.68, a paired difference of -0.45 points with a 95% bootstrap interval of $[-2.06, 0.99]$ (10,000 paired item-level resamples, seed 20260902). A separately executed run on the same items adds the quantized exact-cache arm: full-prefix Q8 retains 56.438 MiB/document at -3.78 points $[-9.12, 0.32]$, so the frozen policy is $5.84\times$ smaller than that arm with a 3.41-point higher mean F1. This is a selected trade-off point among the evaluated split-replay policies, chosen on this same panel and not selection-corrected; it is not a statistical-equivalence claim. Timing is adverse: full-prefix Q16 reaches the first token in 0.163 s against about 0.673 s and has higher generated-token throughput (13.38 versus 7.15–7.19 tokens/s), while TPOT is within 2.5% of its 54.11 ms value, so the supported contribution is capacity reduction with an explicit online cost, not TTFT acceleration. Separately, sharing one document's KV rather than copying it per request lowers the post-priming allocator delta from 4.901 to 2.229 GiB final (54.5%) and 4.920 to 2.843 GiB peak (42.2%) at $N = 32$, against a full-copy arm that is a controlled ownership factor rather than an optimized baseline and reproduces paged-prefix block sharing, which our audited fork ties on both endpoints. The audit exposes a historical alias that output and terminal-state equality miss but a persistent-base content invariant also catches, so its increment is transition-time localization, not exclusive detection. We do not claim lower total process memory, measured recall, edge-device performance, or broad quality preservation.

1 INTRODUCTION

Repeated queries over stable documents are common in document QA, agents, tree search, and best-of-N decoding. Reusing a document prefix avoids repeated prefill, but exact caching retains per-document state at every model layer, which on a memory-constrained deployment competes with model weights, request workspace, and other resident documents. The question is therefore capacity-first: how much reusable document state must stay resident, and what latency and quality cost buys a smaller footprint (Section 3)?

Dense recomputation and exact prefix caching are the two endpoints: the former stores no document state but repeatedly processes the prefix, the latter gives low warm TTFT but retains the complete prefix cache. CoMem instead writes a document to an intermediate depth and reuses the resulting per-token residual (Liu et al., 2026a;b). Hybrid LLMs make this harder: lower full-attention layers also need KV, lower linear layers need convolution and recurrent state, and those mutable states cannot be shared blindly across requests (Yang et al., 2025; Lenz et al., 2025).

We introduce Q-COMEM, a quantized hybrid split-replay design (Figure 1). Its offline Write persists the split residual and all lower-layer state a future query needs under state-type-aware low-bit packing; its online Read dequantizes that entry, shares only immutable document tensors, creates request-local mutable state, and reconstructs the upper suffix, spending computation to save retained bytes rather than trying to dominate exact caching on TTFT. Because equal outputs can still hide a mutable-state alias, we pair the design with FORKAUDIT, a phase-aware ownership trace over immutability, copy-on-write (COW), recurrent rebinding, and selected call provenance, evaluated on a full-prefix BF16 configuration rather than on the quantized Read path. Our contributions follow the memory–latency–quality logic of the rest of the paper:

- **Whole-entry accounting for a hybrid split-replay entry.** Q-COMEM applies group-wise asymmetric quantization and per-layer, per-state-type width assignment — neither claimed as new (Section 2) — to an entry those methods do not address: the depth-j residual *together with* the lower full-attention KV and convolution/recurrent state a hybrid backbone needs to replay a later query. CoMem retains only the residual, unquantized in BF16, so counting it alone understates a hybrid deployment; every number we report uses the whole-entry denominator of Section 4.1.
- **Memory, latency, and quality measured separately.** On 60 archived Qasper/2WikiMQA items, the frozen Q4/Q4/Q8 point is $14.10\times$ smaller than the full-prefix reference and changes mean F1 against that same reference by -0.45 points $[-2.06, 0.99]$ (Sections 5.2 and 5.4); a separate execution on an eight-item subset reports TTFT, TPOT, and throughput with the adverse TTFT trade-off visible (Section 5.3), and a separately executed run on the same items supplies the quantized exact-cache arm, $5.84\times$ larger in retained state and 3.41 F1 points lower than the frozen point. Both tables keep retained payload separate from allocator and process memory and leave unmeasured Recall cells blank.
- **Ownership-safe reuse, audited off the quantized path.** The seven-target FORKAUDIT contract couples semantic relations to phase-indexed storage evidence across 96 ownership configurations, all full-prefix BF16 KV with no split depth and no quantization; the quantized Read path is not audited. A historical alias corrupts the persistent base while output and terminal request state stay exact, but a persistent-base content invariant also catches it, so the increment is transition-time owner/layer/family localization, not exclusive detection. At $N = 32$ shared-document KV cuts the post-priming allocator delta by 54.5% final and 42.2% peak against a full-copy ownership control, matching this paper's own paged-prefix baseline (Section 5.5).

The timing execution uses an eight-item subset, the operating-point run a separate execution, and the ownership study a third system entirely (Section 5.1); "Store" is retained tensor payload, not total VRAM, process memory, or admitted serving capacity (Section 5.6).

2 RELATED WORK

Q-COMEM combines depth reuse, exact prefix caching, cache compression, hybrid state, and relation-based testing around one objective: reduce the complete retained state of a hybrid split-replay entry while making request-local ownership auditable.

Split-depth reuse. CoMem persists one depth-j residual per document token and executes the upper suffix for later queries (Liu et al., 2026a;b); we build on that interface rather than claiming a

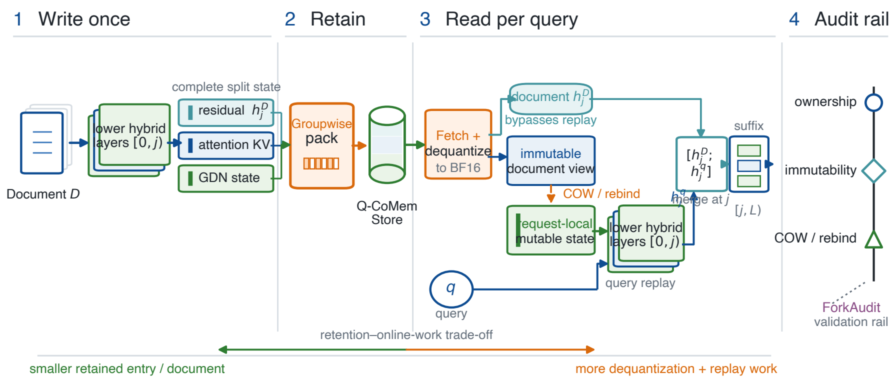

Figure 1: Q-CoMem protocol geometry. Write runs a document once to split depth j and packs the boundary residual, lower full-attention KV, and lower GDN convolution/recurrent state. Read dequantizes that entry, exposes immutable document tensors, creates request-local mutable state through COW/rebinding, and replays the query through $[0, j)$. The narrow ForkAudit rail validates the ownership boundary and is not an execution stage; it ran only on a full-prefix BF16 configuration with no split depth and no quantization, so the packed Read path drawn here is unaudited.

new split-replay primitive, and add the problem of quantizing the residual *and* the lower KV/recurrent state a hybrid model needs under an explicit storage denominator.

Paged and prefix-aware inference. PagedAttention, Prompt Cache, RadixAttention, ChunkAttention, Hydragen, and Preble establish block sharing, prefix reuse, shared-prefix computation, and scheduling (Kwon et al., 2023; Gim et al., 2024; Zheng et al., 2024; Ye et al., 2024; Juravsky et al., 2024; Srivatsa et al., 2025); they are the low-TTFT reference point, and Q-CoMem trades suffix work for a smaller entry.

Hybrid state and cache management. Gated Delta Networks maintain mutable compact state (Yang et al., 2025); Jamba, Marconi, and HYPIC address hybrid architectures or reusable hybrid prefixes (Lenz et al., 2025; Pan et al., 2025; Liu et al., 2026c) and store that hybrid state unquantized. Q-CoMem differs by jointly accounting for the split residual and the lower hybrid state under one denominator, and ForkAudit validates the immutable-document/mutable-request boundary rather than replacing these cache policies; Appendix Table 19 separates published numbers from our controls.

KV and state quantization. Low-bit storage of reusable state is established prior art and Q-CoMem claims none of its mechanisms. The packer of Section 4.2 is the standard asymmetric affine quantizer (Jacob et al., 2018) applied group-wise — the form FlexGen already used for weights and the KV cache as 4-bit codes over 64-element groups with per-group min/max metadata (Sheng et al., 2023) — and KIVI, KVQuant, and XQuant fix the grouping axis or the tensor quantized (Liu et al., 2024; Hooper et al., 2024; Tomar et al., 2025). Per-layer width assignment is likewise prior art (Li et al., 2025a; Tao et al., 2025; Yang et al., 2024): KVTuner searches layer-wise mixed-precision KV bit pairs offline and AsymKV gives keys and values separate per-layer widths, while Palu, SqueezeAttention, and HeadKV allocate rank or budget rather than bits (Chang et al., 2025; Wang et al., 2025; Fu et al., 2025) and Quamba, Quamba2, and MambaQuant quantize selective-SSM activations and per-state-group recurrent inputs, including for a hybrid backbone (Chiang et al., 2025b;a; Yue et al., 2025). Our bit vector $\mathbf{b}$ is an application of an established design surface, not a new selection method. What it does not cover is the object quantized: each method compresses one homogeneous cache along a single execution path, whereas a hybrid split-replay entry is heterogeneous, and we report Store for the whole entry (Section 4.1) rather than the residual alone, which CoMem retains unquantized in BF16 (Liu et al., 2026a).

Evaluation and metamorphic testing. SCBench emphasizes cache lifecycles (Li et al., 2025b); metamorphic testing, MT-DLComp, and CRADLE motivate preserved relations and independent checks (Chen et al., 2018; Xiao et al., 2022; Pham et al., 2019); DNNMem separates memory denominators (Gao et al., 2020). Appendix Table 14 gives our boundary.

3 Motivation

Retained document state is the controllable term. A deployment must fit model weights, shared adapters, active workspace, retained document entries, and runtime state at once:

$$M_{\text{device}} \geq M_{\text{weights}} + M_{\text{shared}} + M_{\text{active}} + \sum_{D \in \mathcal{H}} S(D) + M_{\text{runtime}}, \tag{1}$$

where $\mathcal{H}$ is the hot document set. Weights and runtime are fixed once a model and stack are chosen and workspace is set by the request shape, so the retained term is the only one that grows with how many documents stay warm. Q-CoMem targets $S(D)$: it does not compress weights or promise a smaller workspace, reports allocator measurements under a separate denominator (Section 5.1), and leaves process/NVML memory unmeasured.

Neither endpoint gives a small $S(D)$ at usable warm latency. Dense recomputation sets $S(D) = 0$ but pays full prefill on every query. Exact prefix caching is the low-TTFT reference point but retains per-document state at every layer: on our stack 136.235 MiB/document (Section 5.1, Table 1). Because Eq. 1 multiplies that by $|\mathcal{H}|$, the endpoint best for warm latency is worst for capacity. Split replay at depth j sits between them, and quantizing what it retains moves it further toward the capacity end at a suffix-computation cost Section 5.3 measures.

In a hybrid model the entry is not just larger — it is mutable. Replaying a query through the lower layers needs more than h_j^D: lower full-attention layers need their KV and lower linear layers need convolution and recurrent state, so an honest denominator must count the whole entry. And those lower-layer states are mutated in place by the kernels that consume them, so sharing one entry across concurrent requests is an ownership question output comparison cannot answer: we observe a historical configuration whose tokens and terminal request state stay exact while a persistent base is corrupted, which a persistent-base content invariant also catches, but only at final capture and without naming the owner, layer, or state family responsible (Section 5.5). These fix the method that follows — quantize the complete entry (Sections 4.1–4.2), fork it under an explicit ownership discipline (Section 4.3), and validate that discipline with storage evidence rather than output equality (Section 4.4).

4 Methodology

4.1 Retained state decomposition

For split depth j the reusable hybrid state comprises the document boundary residual h_j^D, lower-layer full-attention KV, and lower-layer convolution/recurrent state; with quantizer Q_b, the retained payload of Eq. 1 is

$$S_{\text{QCoMem}}(D; j, \mathbf{b}) = |Q_{b_h}(h_j^D)| + \sum_{\ell < j, \ell \in \mathcal{A}} |Q_{b_\ell}(K_\ell^D, V_\ell^D)| + \sum_{\ell < j, \ell \in \mathcal{G}} |Q_{b_\ell}(C_\ell^D, G_\ell^D)|, \tag{2}$$

where $\mathcal{A}$ and $\mathcal{G}$ index full-attention and recurrent layers. Counting only h_j^D would understate the deployed state. The bit vector $\mathbf{b}$ is assigned per state type and per layer, and every policy we evaluate is one frozen setting of it.

4.2 Offline Write: packing the complete entry

The document runs through layers $[0, j)$; we retain h_j^D and the lower attention and recurrent cache leaves a later query needs, partitioning each floating tensor into groups of 64 values. For a b-bit group with minimum m and maximum u, we store

$$s = (u - m)/(2^b - 1), \qquad q = \text{clip}\big(\text{round}((x - m)/s), 0, 2^b - 1\big), \tag{3}$$

with packed unsigned values and BF16 scale/bias metadata. Q16 denotes the unpacked reference: the residual is cast to BF16, while cache leaves are kept in the dtype the stack produces, which for GDN recurrent state is FP32 (Section 5.1). The evaluated Q4/Q4/Q8 point assigns Q4 to the residual and lower full-attention state and Q8 to lower GDN convolution/recurrent state; Figure 2 separates these retained-state widths from the unchanged BF16 model layers and forward computation.

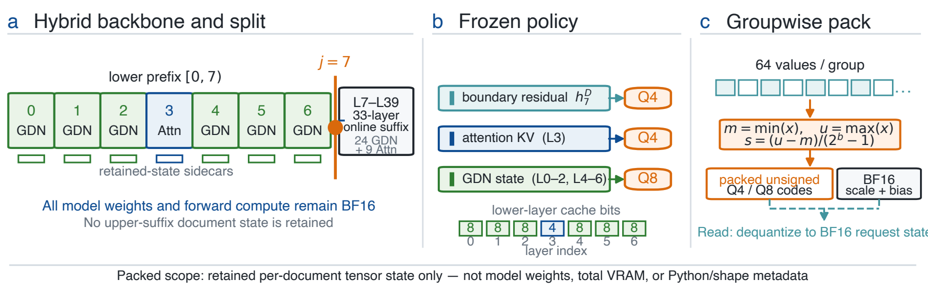

Figure 2: Quantization scope and frozen Q4/Q4/Q8 policy. (a) At split $j = 7$ the retained entry holds the boundary residual and lower-layer (L0–L6) state sidecars; weights and forward computation stay BF16, and L7–L39 state is reconstructed on Read. (b) The frozen point packs h_7^D and L3 attention KV at Q4 and the L0–L2/L4–L6 GDN convolution/recurrent state at Q8. (c) Each 64-value group stores packed unsigned codes plus BF16 scale and bias. Q16 denotes the unpacked reference: BF16 for the residual and lower attention KV, the stack's native FP32 for GDN recurrent state. The scope is retained tensor payload, not weights or total VRAM.

The reported per-layer policy uses a Q4 residual and active-layer widths $[8, 8, 4, 4, 8, 8, 8]$ selected on a disjoint four-item calibration set; we treat it as one frozen operating point, not a claim that layer-wise selection is optimal.

4.3 Online Read: ownership-safe fork and suffix replay

For each query, the packed entry is dequantized into a logical document view. For resident requests $r \in \{1, \ldots, N\}$, the logical fork of that view is $\mathcal{F}_r(D) = \big(\{K_\ell^D, K_\ell^r\}_{\ell \in \mathcal{A}}, \{G_\ell^D, G_\ell^r\}_{\ell \in \mathcal{G}}\big)$, where document state K_ℓ^D, G_ℓ^D is immutable and may be shared, whereas append reservations K_ℓ^r and mutable recurrent state G_ℓ^r are request-local. A borrowed recurrent base is read-only at setup and must rebind to private storage when the registered transition first mutates it; a partial KV tail is copied before append. This discipline is what ForkAudit audits; the Transformers implementation behind Tables 1 and 2 instead materializes a full private copy of the dequantized entry per query, so it shares nothing and exercises neither borrowing nor copy-on-write. Query tokens then traverse the lower layers using request-local state; document and query residual chunks — kept distinct, as the Qwen3.5 recurrence requires — seed the suffix, executed through layers $[j, L)$ before greedy decoding.

4.4 Fail-closed validation of the ownership contract

ForkAudit validates the ownership discipline of Section 4.3 with seven targets: frozen identity, prefix immutability, private ownership, tail-safe COW, bounded host-side call provenance, cross-arm semantic equivalence, and cross-N prefix consistency. It runs on a full-prefix BF16 configuration with no split depth and no quantization (Section 5.5), so it validates that discipline and not the dequantize-then-fork Read path; the obligations specific to a packed entry — dequantized-view immutability, residual-chunk binding, packed-entry lifetime — are untested. For case specification Σ, execution trace τ, mandatory slots $M_i(\Sigma)$, and replay predicates Φ_i,

$$\mathrm{Pass}_i(\tau) \iff [\mathrm{Coverage}_i(\tau) = \mathrm{complete}] \wedge \mathrm{Bind}_\Sigma(\tau) \wedge \bigwedge_{\phi \in \Phi_i} \phi(\tau), \tag{4}$$

where completeness quantifies over $M_i(\Sigma)$ — every mandatory slot present, unique, unmodified — and Bind_Σ requires each slot to resolve to the live tensor its receipt names, so coverage is separate from verdict: a missing, duplicated, modified, or unbound mandatory receipt cannot silently pass. Targets 2–5 are pre-output storage/call obligations; targets 6–7 compare registered semantic observables and cannot substitute for ownership evidence. The workflow freezes the run, captures setup/transition/final digests, ranges, calls, semantics, and allocator snapshots, then replays every applicable predicate (Appendix E, Table 24). It is an offline, non-adversarial regression contract, not security attestation: capture/replay, the mandatory-slot schema, slot-to-live-tensor binding, and framework storage semantics form the trusted computing base (TCB), and a pass establishes only that the registered predicates held at captured points under it, covering neither coherent omission/fabrication and transient writes between observations nor compiled GDN, ATen/CUDA, or device/driver execution.

5 Experiments

Sections 5.2–5.4 report capacity, its online cost, and answer quality; Section 5.5 reports whether the ownership discipline of Section 4.3 holds under storage evidence.

5.1 Setup and memory denominators

Quality cohort. The broadest Store–F1 panel is an archival, raw-first validation run on the stack of Table 2. It contains 60 paired items — Qasper and 2WikiMQA source indices 6–35, 30 per dataset — under official LongBench-v1 prompts and a 4,096-token input cap (Bai et al., 2024a; Dasigi et al., 2021; Ho et al., 2020), comparing the six policies of Table 1. Four calibration prompts (indices 4–5 of each dataset) precede and are disjoint from validation; the archived shards provide Store, predictions, references, and LongBench F1, but not timing or Recall. A second, separately executed *operating-point run* covers the same 60 items on the same stack with one warm-up, three repeats, dataset-length generation, and seed 20260903; it records TTFT and TPOT and adds a quantized exact cache (full-prefix Q8, the same codec applied to the unsplit entry) and a $j = 13$ split control. The two executions occupy separate blocks of Table 1, are never pooled, and each is differenced against its own full-prefix arm.

Latency execution. A separate registered systems execution uses the same stack on eight items (Qasper and 2WikiMQA source indices 6–9), a subset of the quality cohort, not an independent replication; the two tables are not pooled. Table 2 states its geometry, configurations, and metrics; Recall was not recorded.

Memory denominators. In both deployment tables “Store” is the physical byte-range union owned by one retained entry, broken into stored components in Appendix C; on the eight documents both tables contain the two implementations agree to the byte at all five retained configurations, so the whole 136.235-versus-140.34 and 9.661-versus-10.01 difference is cohort and statistic — a 60-item mean against a median over documents 3.8% longer — with 0.000 MiB left to the accounting, and every ratio below carries its cohort and statistic. Neither denominator includes non-tensor/Python metadata, allocator pools, process/NVML memory, active workspace, or admission capacity, and the full-prefix reference retains GDN recurrent state in FP32 as the unmodified stack produces it; the allocator result in Section 5.5 uses a third denominator, and every quality difference below uses the full-prefix arm of its own execution (Appendix B).

Executed systems. Tables 1 and 2 are Transformers-only executions of packed split-replay entries, whereas Table 18 and every ForkAudit verdict come from a different system — vLLM 0.26 plus Transformers, full-prefix BF16 KV in 128-token pages, no split depth, no quantization, PG-19 — so no ForkAudit verdict here covers those packed entries.

5.2 Memory: retained state per document

Moving from full-prefix Q16 to split Q16 first reduces mean Store from 136.235 to 34.683 MiB/document ($3.93\times$, 60-item mean). Quantizing the complete split state with the frozen Q4/Q4/Q8 policy reduces it further to 9.661 MiB/document, $14.10\times$ smaller than the full-prefix reference. That reference is what the unmodified stack retains, in its native dtypes, and 30.000 MiB/document of it is FP32 GDN recurrent state in excess of a BF16 encoding, so part of the reduction is dtype narrowing

Table 1: Retained-state–quality trade-off on the 60-item validation cohort (30 Qasper, 30 2WikiMQA) in two separate executions, blocked and never pooled, each block differenced against its own full-prefix row. The operating-point run re-executes the same items with three repeats, records TTFT, and adds the quantized exact-cache arm and a $j = 13$ split control.

Retained state	Store (MiB/doc)	vs. prefix	TTFT (s)	F1 (0–100)	Recall	ΔF1 vs. full-prefix [95% CI]
Archival quality run (seed 20260902; no timing recorded)						
No-cache dense recompute	–	–		54.29		-0.39 [-4.66, 2.60]
Full-prefix, native dtype	136.235	1.00×		54.68		0.00 [0.00, 0.00]
Q-COMEM split Q16	34.683	3.93×		54.62		-0.06 [-0.18, 0.00]
Q-COMEM frozen Q4/Q4/Q8	9.661	14.10×		54.24		-0.45 [-2.06, 0.99]
Q-COMEM same-memory mixed	9.395	14.50×		53.36		-1.32 [-4.33, 0.92]
Q-COMEM aggressive mixed	7.536	18.08×		49.19		-5.50 [-11.58, -0.20]
Operating-point run (seed 20260903; separate execution on the same 60 items)						
No-cache dense recompute	0.000	–	0.636	54.18		-0.49 [-4.57, 2.55]
Full-prefix, native dtype	136.235	1.00×	0.181	54.67		0.00 [0.00, 0.00]
Full-prefix Q8	56.438	2.41×	0.182	50.89		-3.78 [-9.12, 0.32]
Q-COMEM frozen Q4/Q4/Q8, $j = 13$	16.101	8.46×	0.582	51.27		-3.40 [-8.67, 0.52]
Q-COMEM frozen Q4/Q4/Q8, $j = 7$	9.661	14.10×	0.656	54.30		-0.37 [-1.97, 1.08]

Store and F1 are 60-item means; within each block every column, compression included, uses that block's full-prefix arm as the unmodified stack retains it, in its native dtypes. 30.000 MiB/document of that 136.235 MiB is FP32 GDN recurrent state in excess of a BF16 encoding, as is 6.000 MiB of the split row's 34.683 MiB, so the all-BF16 column reads $106.235/28.683/9.661/9.395/7.536$ MiB/document ($1.00/3.7038/10.9965/11.3073/14.0969\times$); normalised, the split-to-frozen step is $2.97\times$ rather than the $3.59\times$ the printed column shows (Appendix C prints every component byte count, the identity each row satisfies, and why the printed step reads above the Eq. 3 ceiling). Intervals are paired 10,000-resample item-level bootstrap percentile intervals against that block's full-prefix arm (seeds 20260902/20260903) on unrounded means; none was computed between two non-reference arms, an interval crossing zero is not statistical equivalence, and the headline policy was selected without multiplicity adjustment (Section 5.4 gives the catastrophic-regression counts and the selection caveat). Appendix B gives the method and all 42 archival intervals; blank cells are unmeasured metrics, not zeros.

rather than the Eq. 3 packing step; against an all-BF16 reference of 106.235 MiB/document the two ratios become $3.7038\times$ and $10.9965\times$. Both are exact re-analyses of the same archived rows on 60/60 items — the physical saving on this stack, and the like-for-like packing saving; Appendix C prints the per-component bytes behind each.

The operating-point run supplies the arm the archival panel lacks: full-prefix Q8, an exact cache packed with the same codec, retains 56.438 MiB/document, $2.41\times$ below the native-dtype reference but $5.84\times$ above the frozen point. Uniform Q4 and the aggressive mixed policy go further still (Table 2), but Section 5.4 shows why neither is the headline point. Store is retained tensor payload only; the separate allocator result in Section 5.5 is not additive with it.

5.3 Latency: TTFT, TPOT, and throughput

The separate timing characterization is not a speedup over exact caching (Table 2). Full-prefix Q16 records 0.163 s TTFT against about 0.673–0.674 s for the Q-COMEM rows, which dequantize and rebuild the suffix; their TPOT is within 2.5% of full-prefix (54.97–55.44 versus 54.11 ms), but median generated-token throughput falls from 13.38 to 7.15–7.19 tokens/s. The operating-point run sharpens this on all 60 items: full-prefix Q16 and Q8 reach the first token in 0.181 and 0.182 s, dense recomputation in 0.636 s, and the frozen $j = 7$ point in 0.656 s, so Q-COMEM is slower to the first token than honest dense recomputation as well as than exact reuse, while TPOT stays between 54.4 and 56.0 ms at every arm. The cost is concentrated in the first token, so the supported result is a retained-capacity trade-off with an explicit online cost and no latency advantage over either endpoint.

5.4 Answer quality

Every quality difference uses the full-prefix arm of its own execution, as the Store and compression columns do. The frozen Q4/Q4/Q8 point has mean F1 54.24 versus 54.68, a paired difference of -0.45 points $[-2.06, 0.99]$ (10,000 paired item-level resamples, seed 20260902), consistent in sign across both datasets (-0.34 Qasper, -0.56 2WikiMQA); the two steps compound rather than cancel, splitting costing -0.06 points and quantizing the split state a further -0.39, and no item crosses the registered catastrophic threshold under either arm (0/60 against dense and against full-prefix). The headline policy was selected among six on this panel, so the interval is not selection-corrected and proves neither equivalence nor noninferiority (Appendix B).

Table 2: Capacity–latency–quality trade-off from a separate execution on the registered eight-item Qasper/2WikiMQA slice (source indices 6–9 of each dataset), a subset of the 60-item panel rather than an independent replication. All rows use Qwen3.5-35B-A3B, PyTorch 2.11.0+cu129, Transformers 5.14.1, one H20-3e per item, a 4,096-token cap, greedy decoding, at most 32 generated tokens, and three timing repeats.

Dataset: Qasper (4 items) + 2WikiMQA (4 items)							
Retained state	Store (MiB/doc)	Reduction	TTFT (s)	TPOT (ms)	tok/s	F1 (0–100)	Recall
No-cache dense recompute	0.00	–	0.649	648.75	1.36	39.42	
Full-prefix, native dtype	140.34	reference	0.163	54.11	13.38	39.14	
Q-CoMem Q8	15.89	88.68%	0.673	55.44	7.15	39.14	
Q-CoMem Q4/Q4/Q8	10.01	92.87%	0.674	54.97	7.19	39.01	
Q-CoMem per-layer mixed	9.74	93.06%	0.673	55.07	7.19	39.11	
Q-CoMem Q4	8.41	94.01%	0.674	55.02	7.19	36.09	

Store, timing, and throughput columns are medians over the eight documents, whereas the 60-item panel reports means over a cohort with 3.8% shorter documents, which is why its full-prefix figure is 136.235 rather than 140.34 MiB. Store is the byte-range union owned by one retained entry and excludes non-tensor/Python metadata, allocator pools, process/NVML memory, workspace, and admission capacity. The reference is the entry in the stack's native dtypes: 42.8% of its 140.34 MiB is FP32 GDN recurrent state, so part of every reduction is dtype narrowing rather than packing. F1 is the eight-workload mean and this execution is not pooled with the 60-item panel. The per-layer policy uses a Q4 residual and layer bits $[8, 8, 4, 4, 8, 8, 8]$ from a disjoint four-item calibration set; its mean-F1 delta is -0.022, Q8's is 0.000. The all-Q4 row is the aggressive failure endpoint, and Recall was not measured.

The operating-point run measures what an exact cache costs once quantized with the same codec. Against that run's own full-prefix reference of 54.668, full-prefix Q8 scores 50.885 (-3.78 $[-9.12, 0.32]$) and the frozen $j = 7$ point 54.297 (-0.37 $[-1.97, 1.08]$), so the frozen point is both $5.84\times$ smaller than the strongest quantized exact cache we measured and 3.41 points higher in mean F1; we computed no paired interval between those two arms and their intervals against the shared reference overlap, so the retained-state ordering is the firmer of the two. The $j = 13$ control loses 3.40 points $[-8.67, 0.52]$, about nine times that loss, fixing the split depth at $j = 7$ for this policy — a control on one parameter, and we report nothing about depths this cohort did not measure. The frozen point's -0.37 $[-1.97, 1.08]$ also reproduces the archival cohort's -0.45 $[-2.06, 0.99]$ against the same reference arm in an independent run.

The two mixed policies are ablations, not improvements: the same-memory per-layer policy is named for its calibration budget, not measured parity, saving 2.75% Store while its mean-F1 difference is more negative and one of 60 items crosses the threshold; the aggressive policy reaches $18.08\times$ with its interval below zero and five such items, and uniform Q4 loses 3.047 F1 points on the eight-item execution. We therefore use frozen Q4/Q4/Q8 at $j = 7$ as the headline point. Recall was not recorded in any cohort, its table cells remain blank, and we make no recall claim.

5.5 Ownership and correctness validation

Scope, cohort, and procedure. The primary factorial runs full-prefix BF16 KV on a vLLM-plus-Transformers stack with no split depth and no dequantization step, and so audits the ownership discipline of Section 4.3, not the quantized Read path of Tables 1 and 2. Three executions produce the results below, each count naming its own: a *primary rerun* over the 96 frozen configurations of Table 5 (one Qwen3.5-35B-A3B stack (Qwen Team, 2026a;b), 8 H20-3e ranks $\times$ *resident counts* $N \in \{1, 8, 32\}$ $\times$ four KV-by-GDN cells, one PG-19 book (Rae et al., 2020) per rank; Figure 3), a *dispatch rerun* for host-side call provenance, and a *live-binding rerun* with a stronger hook on one cell. Appendix A archives all three and defines their shard verdicts; Tables 8–9 give per-target coverage, verdicts, and exclusions.

What it found. All seven targets have complete mandatory-trace coverage and pass at their declared fixed-stack scope: every registered semantic observable matches exactly across the 96 configurations, and completed requests in both setup arms rebind at the registered boundary to request-private tensors disjoint from the bases and from completed peers. The live-binding rerun is scientifically valid on all eight ranks, but its stronger owner-row binding covers only the selected $N = 8$ shared-document/materialized-GDN cell, leaving other cells and the honest-process assumption outside it (Appendix E, Table 6).

Allocator endpoints under a controlled ownership factor. At $N = 32$ with materialized GDN setup, shared-document rather than full-copy KV cuts the final allocated delta from 4.901 to $2.229\,\mathrm{GiB}$ (54.5%) and the peak from 4.920 to $2.843\,\mathrm{GiB}$ (42.2%) above the frozen post-priming baseline, registered observables equal. The full-copy arm is a controlled ownership factor, not an optimized baseline: no serving engine copies a shared prefix per request, and the saving is block sharing (Kwon et al., 2023). Against Table 18's paged-prefix baseline, the closest same-stack reference, our fork ties on both endpoints and is worse on generation increment (1.950 versus $0.019\,\mathrm{GiB}$), buying auditable request-local state, not fewer allocator bytes.

Falsification beyond output equality. Every paired mutant is rejected at its frozen gate, yet token equality catches 0 of 5 semantics-completing mutants and exact logits 1 of 5 (Table 11; Table 12's designer–executor-separated campaign is a different cohort, not the source of that fraction). In a no-injection defect, output and terminal request state stay exact in 8/8 cells under persistent-base corruption that Table 17's content invariant also catches on all eight pre-fix cells, so the increment is authenticated transition-time owner/layer/family localization, not exclusive detection. These and Appendix H's supporting cohorts, Falcon-H1 (Zuo et al., 2025) included, are fixed-case comparisons against registered observables establishing sensitivity and localization, not a head-to-head detection study or a population error rate (Appendix F, Tables 10–12).

5.6 Limitations and scope

The quality panel is 60 Qasper/2WikiMQA items on one Qwen3.5/H20 stack, the latency panel a separate execution on its eight-item subset, the operating-point run a third on the same items; none replicates another, none is pooled, and they support bounded point estimates, not broad preservation. The archival run is raw-first evidence — hashes, predictions, F1, intervals, and Store accounting all replay — but its launcher used a mutable code directory, froze neither a full-source nor a model-weight ledger, and left parts of its toolchain unarchived, so it is not source-frozen regeneration (Appendix A); it has no timing and no cohort measured Recall. H20 is not an edge device, so local or edge use is motivation (Section 3), not an evaluated claim; the calibrated per-layer row is one operating point from four disjoint items; the split depth is validated against one alternative ($j = 13$), not swept; TTFT is substantially higher than full-prefix reuse and, on the operating-point run, than dense recomputation too; and we do not evaluate continuous batching, QPS, admission, eviction, or scheduler effects.

Store, the entry-owned physical byte-range union, excludes non-tensor/Python metadata, allocator pools, active workspace, process/NVML memory, and weights; reducing it does not by itself prove lower peak VRAM or serving capacity, and the separate 54.5%/42.2% allocator result is a post-priming delta against a full-copy control, neither additive with Store nor evidence of higher admitted capacity.

FORKAUDIT has not been run on the quantized Read path: every ownership cell uses full-prefix BF16 KV with no split depth or dequantization, whereas Tables 1 and 2 measure a $j = 7$ packed Q4/Q4/Q8 path whose Read step materializes a full private copy rather than borrowing it, so ownership there transfers by design argument only. Within the TCB of Section 4.4 it upgrades semantic equality to phase-indexed ownership evidence, but its selected-cell binding is a same-process check: other cells and coordinates, framework/IPC semantics, compiled GDN, ATen/CUDA, and device/driver execution stay unattested, and constructed controls plus one historical mechanism show fixed-case sensitivity, not population error rates (Appendix I).

6 Conclusion

Q-COMEM jointly packs the document residual and the lower KV/recurrent state a later query needs. On 60 items, frozen Q4/Q4/Q8 at $j = 7$ is $14.10\times$ smaller than the full-prefix reference ($10.9965\times$ all-BF16) and $5.84\times$ smaller than a quantized exact cache at 3.41 F1 points better, at -0.45 $[-2.06, 0.99]$ against full-prefix Q16. The cost is explicit: exact reuse is much faster in TTFT and higher in throughput, dense recomputation reaches the first token sooner, TPOT within 2.5%. FORKAUDIT localizes request-local ownership violations on a full-prefix BF16 configuration, and the claim is capacity-first: smaller retained state per document at bounded measured quality, not Recall, total VRAM, or faster inference.

Ethics Statement

This work introduces no human-subject data collection. It uses public PG-19 text for the trace-validation protocols and a fixed 60-item Qasper/2WikiMQA cohort, whose eight-item subset carries the latency measurements. It does not assess model safety, bias, privacy, or downstream deployment quality. Memory efficiency can lower deployment cost but may also increase the scale of model use; those broader effects are outside this controlled systems study.

Use of Large Language Models

The authors used LLM-based assistance throughout the preparation of this manuscript: drafting and revising prose, LaTeX, and tables; triaging related work; generating internal critiques and revision plans that the authors adjudicated; and drafting experiment specifications, including the registered response plan that selected which additional runs this revision executed. Every experiment was run on author-controlled infrastructure, and the authors verified each numerical claim against the archived artifacts and take full responsibility for the content, including any remaining errors.

Reproducibility Statement

Claims map to registered evidence identifiers. The 60-item table maps to a raw-first archival package that rechecks 66 mirrored files, 48 shards, 360 item-level F1 values, 24 paired-bootstrap intervals, and Store accounting; the 18 further intervals of Appendix B are recomputed from those same archived per-item values. Table 1's operating-point block comes from a separate registered execution (one warm-up, three repeats, seed 20260903) recorded under its own run identifier; it is not part of that archival package and the two are never pooled. The latency table is hash-bound to its eight workload shards and aggregate, manifest-first replay covers the primary ownership factorial and named supporting cohorts, and dispatch and live-binding results bind frozen archives to post-run ledgers, terminal manifests/trees, and audit mirrors. Appendix A reports replay counts, costs, non-reexecuted boundaries, and the unarchived parts of the 60-item toolchain.

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A Detailed Reproducibility and Artifact Scope

The 60-item quality package (archive alias QP-60; manifest SHA-256 `24ac6952c6235bca6342171a9769f9da2d0e50d021e32c6ed453d931fd6c9952`) preserves 66 authoritative mirrored files from its recorded execution, including 48 configuration-by-rank shards. Its raw-first verifier recomputes all 360 item-level F1 values, 24 paired-bootstrap intervals, Store component accounting, and the archived aggregate. The data hash and policy receipt are bound and calibration precedes validation, but the original execution did not freeze a complete source or model-weight ledger; offline outcome replay therefore does not imply source-frozen GPU regeneration. The package archives the quantizer/split-replay module and the prompt-and-scoring driver byte-identically at four paths, but not the run driver, the aggregation and bootstrap scripts, the layer-policy search, the deployment accountant, or the launch script; an artifact reviewer can therefore read the code those files call but not the files that produced the table, and they must be released before Table 1 is independently re-executable. The cohort is indexed against the LongBench-v1 release and its official F1 implementation (Bai et al., 2024c;b). The archived receipt binds the prepared data file by content hash, but the upstream dataset revision and evaluation-code commit were not recorded at execution time and cannot be reconstructed after the fact; a re-executor must therefore match the archived data hash rather than resolve an upstream revision.

Manifest-first CPU replay covers the primary-factorial package and named replayable supporting cohorts, not every contextual table. The primary package rechecks 536 artifacts, 96 cells, 288 cross-N comparisons, eight attention rows, and nine fault pairs. The target-gate-suppression package rechecks 18 cases and 24 FP32 sidecars; its disclosed wrapper changes only an inherited execution-identifier source and preserves every candidate byte. The independent-census package rechecks six IPC captures, 1,080 receiver rows, 96,660 relations, and omission/duplication/relabel controls against a preproducer 180-slot expectation. The dual-producer package binds two fresh serial executions of the same frozen producer implementation and four distinct observers, and rechecks exact zero-tolerance equality of the 1,080 observations and 96,660 relation labels per execution. The bounded two-stream replay recomputes four full-logit comparisons and both interval overlaps.

The fresh compiled-dispatch rerun binds a frozen source archive to a passing eight-rank formal aggregate, eight rank receipts, 209,920 attention calls, 635,520 GDN calls, and 224 rejected post-run negative-control decisions. The local mirror preserves the primary and formal aggregates, runtime preflight, raw/scientific ledgers, and both terminal ledgers; it does not duplicate the model view or every raw shard. The receipt closes target 5 only at the declared honest-process fixed-stack boundary: normal launcher return is not a device-side completion attestation, and compiled GDN, underlying ATen/CUDA identity, and driver/device binaries remain outside scope.

The separate live-binding rerun binds a different frozen archive to eight scientifically valid rank shards and a terminally complete formal tree. Its selected-cell hook validates 144 selected rows, 12,960 full live-storage rows, 3,840 direct clone edges, and 24 stable phase artifacts; a separately enumerated schedule counter sees all 96 clean-memory calls with zero selected-context overlaps. The post-run audit also rejects 184 formal controls and verifies 31 bytecode files, 13 descendant directories, and the 1,367-node final tree. This supports only the selected $N = 8$ shared-document/materialized-GDN cell and six preregistered owner-addressed rows per phase under an honest same-process runtime.

Within each selected witness call, the verifier (1) enumerates 60 persistent sources before the builder, (2) records the actual builder's persistent-rooted Torch-dispatch clone edges, (3) exhaustively matches 480 destinations and passes a one-use capability bound to the exact group and verifier, (4) rereads all 540 live tensors against the production serializer at three phases, and (5) publishes hash-bound phase paths for terminal reread. Any count, object, descriptor, lineage, phase-order, or path mismatch aborts before acceptance. Trusted components are hook placement, Python/Torch-dispatch and tensor/storage semantics, and honest process scheduling; group identity, object continuity, lineage cardinality, serializer fields, and publication paths are checked.

The designer–executor-separated package binds the PDF-only author freeze, a lifecycle-amended executor, five clean/mutant pairs, detached clean and pair replays, and 208 terminal files. All five clean gates and expected-first classifications pass; the pre-injection amendment was frozen before all five ranks were rerun in a new output root.

The historical-regression package binds the pre-discovery chronology, frozen pre-fix/repair sources, outcome-blind protocol and resource amendment, eight three-lane rank artifacts, 24 FP32-logit sidecars, and 24 fresh nonces. A fresh candidate-import-free replay of every rank and the aggregate is byte-identical to the archived output.

The expanded numerical-sweep package binds 272 sidecars totaling 140,199,936 bytes and rechecks 20 attention and 24 GDN clean rows plus 44 controls. The Falcon package compares candidate arms against a separately executed official Transformers reference on 8 ranks and detached-replays the aggregate. HYPIC binds 4,404 timing artifacts and hash-replays all 16 Store cells. No package enumerates OS/driver allocations; every conclusion remains bounded to its declared configuration. Appendix J maps exact artifacts and boundaries.

On one Apple-silicon (M4-class) laptop CPU host, three consecutive fixed-order replays without a cache flush took a median 100.146 s (99.248–110.526 s) for the core package and 105.717 s (104.846–119.566 s) for the serial profile that additionally ran five locally complete supporting verifiers; the paired within-profile increment was 5.598 s (5.571–9.039 s). The canonical core distribution has 630 files and 892,284,156 logical bytes (850.95 MiB); 536 raw traces occupy 888,785,811 bytes, or 99.62% of manifest-payload bytes. These are local CPU replay and logical-storage measurements, not H20 capture cost, GPU perturbation, cold-start latency, online inference cost, or a matched uninstrumented delta.

B Complete Interval Inventory and the Reference-Arm Change

Every paired-bootstrap interval that exists for the 60-item panel is listed here, so that none is computed but unreported. The archival analysis computed 24 intervals: six configurations against dense and against split Q16 pooled (12), and six configurations against dense on each dataset (12). *None used full-prefix Q16 as its reference arm*, although the Store column and every compression ratio in Table 1 do. This revision therefore recomputes 18 further intervals against full-prefix Q16 — six configurations $\times$ pooled/Qasper/2WikiMQA — from the same 360 archived item-level F1 values, and Table 1 now displays the six pooled ones. No measurement was repeated: the recomputation reproduces all six published mean-F1 values and all five published Store values exactly (the largest deviation of any recomputed point estimate from the archived one is 6.11×10^{-16} F1 points), and reproduces the archived intervals to within Monte-Carlo noise. The archived analysis records no bootstrap seed, so its endpoints are reproducible only up to that noise; the new intervals use seed 20260902, 10,000 paired item-level resamples of the 60 common $(\mathrm{dataset},\ \mathrm{source\,index})$ keys, and a percentile interval. Across five seeds the frozen row's endpoints move by at most 0.075 points, so the intervals are quoted to two decimals.

Two consequences of the change of arm are worth stating. First, the two steps compound rather than cancel: -0.0585 for splitting plus -0.3870 for quantizing the split state equals the -0.4455 reported against full-prefix, so the previously published -0.39 understated the deployment-relevant loss by about 13% of its value. Second, the frozen policy's registered safety count is unchanged by the change of arm: $0/60$ items below the -50-point catastrophic threshold against dense and $0/60$ against full-prefix.

Why dense is not the per-item reference. The archived rows show identical prefix, document, query and input token counts, identical truncation flags, and identical items, questions and references across all six arms, so the wide dense interval is not an execution-boundary artifact. What the rows do show is that dense recomputation and cached-prefix continuation emit different greedy token sequences on 6 of 60 items, two of them short-answer 2WikiMQA questions whose per-item F1 is nearly binary; the per-item difference standard deviation is 14.51 against 0.45 for full-prefix versus split Q16, which is why the interval is $41\times$ wider. Logits are not archived, so we report the divergence rather than asserting its mechanism. By the same measure Q-CoMem split Q16 tracks full-prefix Q16 more closely ($58/60$ identical token sequences) than dense recomputation does ($54/60$), and on both items where it differs its output is the one that matches dense. The dense row is therefore retained as a no-retention reference, not as a per-item baseline.

Table 3: All 42 paired-bootstrap intervals for the 60-item panel, on the 0–100 F1 scale. The first block is the 18 intervals computed for this revision against full-prefix Q16 (seed 20260902); the second is the 24 intervals in the archival analysis, with the reference arm each one actually used. Rows in the first block with a pooled subset are the ones displayed in Table 1.

Configuration	Subset	ΔF1	95% CI
This revision; reference arm: full-prefix Q16, native dtype (18 intervals)			
No-cache dense recompute	pooled	−0.3934	[−4.6648, +2.6049]
No-cache dense recompute	Qasper	+1.2131	[−0.2339, +3.3305]
No-cache dense recompute	2WikiMQA	−2.0000	[−10.0000, +4.0000]
Full-prefix Q16	pooled	+0.0000	[+0.0000, +0.0000]
Full-prefix Q16	Qasper	+0.0000	[+0.0000, +0.0000]
Full-prefix Q16	2WikiMQA	+0.0000	[+0.0000, +0.0000]
Q-CoMem split Q16	pooled	−0.0585	[−0.1754, +0.0000]
Q-CoMem split Q16	Qasper	−0.1170	[−0.3509, +0.0000]
Q-CoMem split Q16	2WikiMQA	+0.0000	[+0.0000, +0.0000]
Q-CoMem frozen Q4/Q4/Q8	pooled	−0.4455	[−2.0586, +0.9907]
Q-CoMem frozen Q4/Q4/Q8	Qasper	−0.3354	[−3.3818, +2.3932]
Q-CoMem frozen Q4/Q4/Q8	2WikiMQA	−0.5556	[−1.6667, +0.0000]
Q-CoMem same-memory mixed	pooled	−1.3174	[−4.3322, +0.9193]
Q-CoMem same-memory mixed	Qasper	−0.2539	[−3.3148, +2.3720]
Q-CoMem same-memory mixed	2WikiMQA	−2.3810	[−7.1429, +0.0000]
Q-CoMem aggressive mixed	pooled	−5.4961	[−11.5805, −0.1984]
Q-CoMem aggressive mixed	Qasper	−3.9445	[−11.4709, +1.5691]
Q-CoMem aggressive mixed	2WikiMQA	−7.0476	[−17.7143, +1.3333]

Archival analysis; reference arm as listed (24 intervals)

Configuration	Reference arm	Subset	95% CI
No-cache dense recompute	dense	pooled	[+0.0000, +0.0000]
No-cache dense recompute	split Q16	pooled	[−4.6063, +2.6650]
No-cache dense recompute	dense	2WikiMQA	[+0.0000, +0.0000]
No-cache dense recompute	dense	Qasper	[+0.0000, +0.0000]
Full-prefix Q16	dense	pooled	[−2.6066, +4.5088]
Full-prefix Q16	split Q16	pooled	[+0.0000, +0.1754]
Full-prefix Q16	dense	2WikiMQA	[−4.0000, +10.0000]
Full-prefix Q16	dense	Qasper	[−3.3305, +0.2339]
Q-CoMem split Q16	dense	pooled	[−2.6650, +4.4572]
Q-CoMem split Q16	split Q16	pooled	[+0.0000, +0.0000]
Q-CoMem split Q16	dense	2WikiMQA	[−4.0000, +10.0000]
Q-CoMem split Q16	dense	Qasper	[−3.4475, +0.0000]
Q-CoMem frozen Q4/Q4/Q8	dense	pooled	[−3.3045, +4.2800]
Q-CoMem frozen Q4/Q4/Q8	split Q16	pooled	[−2.0365, +1.0756]
Q-CoMem frozen Q4/Q4/Q8	dense	2WikiMQA	[−4.3333, +9.4444]
Q-CoMem frozen Q4/Q4/Q8	dense	Qasper	[−4.6073, +0.7317]

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Table 3 continued.

Configuration	Subset	ΔF1	95% CI
Q-CoMem same-memory mixed	dense	pooled	[−5.2845, +3.7838]
Q-CoMem same-memory mixed	split Q16	pooled	[−4.2793, +0.9775]
Q-CoMem same-memory mixed	dense	2WikiMQA	[−8.4762, +8.6667]
Q-CoMem same-memory mixed	dense	Qasper	[−4.9649, +1.6662]
Q-CoMem aggressive mixed	dense	pooled	[−11.9941, +1.5786]
Q-CoMem aggressive mixed	split Q16	pooled	[−11.5615, −0.1713]
Q-CoMem aggressive mixed	dense	2WikiMQA	[−16.7619, +6.6667]
Q-CoMem aggressive mixed	dense	Qasper	[−12.9395, +1.0689]

C Per-Component Store Accounting and the Eq. 3 Ceiling

Table 1's Store column is a sum of per-component byte counts, and this appendix prints them so that each row can be checked against Eq. 3 without back-derivation. For a component of n values packed at $b < 16$ bits in groups of 64 with BF16 scale and bias, the packed size is

$$\mathrm{bytes}(n, b) = \lceil n/64 \rceil\,(8b + 4),$$

so the format ceilings against a 128-byte BF16 group are $3.5556\times$ at $b = 4$ and $1.8824\times$ at $b = 8$. Every quantized item–policy pair in the archival cohort satisfies this identity exactly (180/180, zero mismatches).

Table 4 gives each component at the cohort mean document length $T = 3807.25$ tokens. Two columns are needed for the unpacked reference: the stack retains Qwen3.5 GDN recurrent state in FP32, so its *native* cost is twice the cost it would carry under an all-BF16 encoding. Every other component is BF16 natively, and $j = 7$ places one full-attention layer (layer 3) and six GDN layers below the split.

Table 4: Per-component retained payload in MiB/document at the 60-item mean document length ($T = 3807.25$ tokens), by stored width. "Native" is the dtype the unmodified stack produces; "BF16 ref." re-expresses GDN recurrent state in BF16. Quantized entries are $\lceil n/64 \rceil (8b + 4)$ bytes.

Component	values n	Native	BF16 ref.	Q8	Q4	Q2
Boundary residual h_j^D	$2048\,T$	14.8721	14.8721	7.9008	4.1828	2.3238
Layer-3 attention K	$512\,T$	3.7180	3.7180	1.9752	1.0457	0.5809
Layer-3 attention V	$512\,T$	3.7180	3.7180	1.9752	1.0457	0.5809
GDN convolution state, per layer	32,768	0.0625	0.0625	0.0332	0.0176	0.0098
GDN recurrent state, per layer	524,288	2.0000	1.0000	0.5312	0.2812	0.1562

Each Table 1 row is a sum of these entries. The full-prefix reference spans all ten attention layers and all thirty GDN layers, $10 \times 2 \times 3.7180 + 30 \times 0.0625 + 30 \times 2.0000 = 136.235$, of which $30 \times 1.0000 = 30.000$ MiB is the FP32 excess; its all-BF16 value is 106.235. Split Q16 retains the residual, layer-3 K and V, and six GDN layers, $14.8721 + 2 \times 3.7180 + 6 \times 0.0625 + 6 \times 2.0000 = 34.683$, of which $6 \times 1.0000 = 6.000$ MiB is the same excess, so its all-BF16 value is 28.683. The frozen policy packs the residual and layer-3 K/V at Q4 and the six GDN layers at Q8, $4.1828 + 2 \times 1.0457 + 6 \times (0.0332 + 0.5312) = 9.661$; the same-memory policy moves layer 2 to Q4, giving 9.395; and the aggressive policy uses Q2 for layer-3 attention and for GDN layers 2, 4 and 6, giving 7.536.

This is what dissolves the apparent format violation in the printed column. Taken as printed, the quantization step reads $34.683/9.661 = 3.590\times$, above the $3.5556\times$ Q4 ceiling. Both numerator and denominator must be stated under one dtype convention: with the split row's 6.000 MiB of FP32 excess removed, the step is $28.683/9.661 = 2.969\times$, inside the ceiling, and the lower-cache-only sub-ratios are $2.5211\times$ (frozen) and $2.6496\times$ (same-memory) against the same $3.5556\times$ bound, with the aggressive policy at $4.1187\times$ against its own $6.4000\times$ Q2 bound. The printed $3.590\times$ is a physical saving on this stack that mixes packing with dtype narrowing; it is not a claim that Eq. 3 compresses beyond its format.

D Protocol Geometry

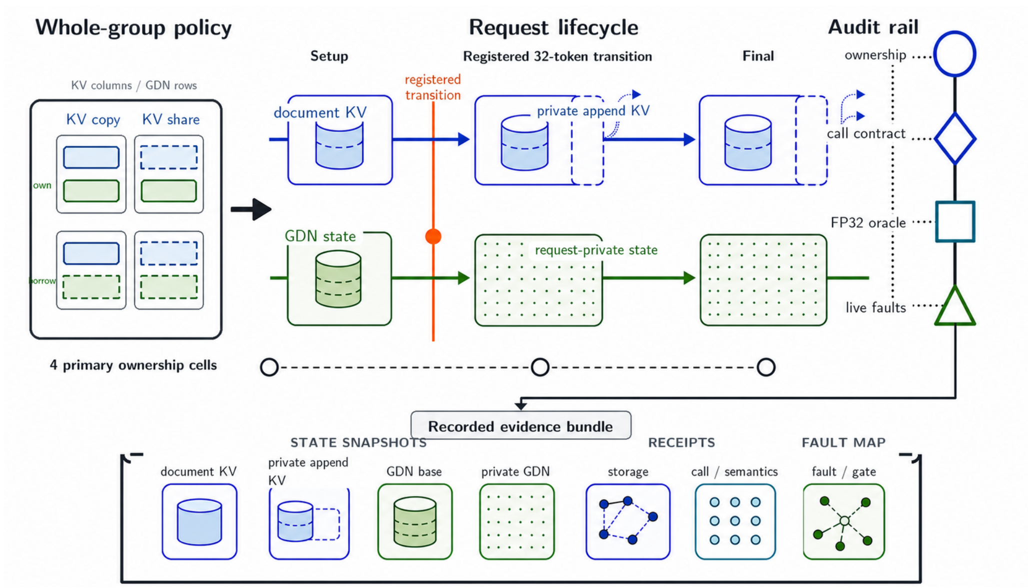

Figure 3: ForkAudit ownership factorial. Four KV-by-GDN setup cells share immutable document state, keep append tails private, and rebind borrowed recurrent state at the registered transition. The audit rail records lifecycle, storage, call/semantic, and fault-to-gate receipts. This schematic details the validation component; it is not experimental evidence.

Table 5: Frozen primary-factorial execution geometry. The eight ranks provide distinct books and query banks, not repeated stochastic measurements.

Component	Frozen value
Model	Qwen3.5-35B-A3B; 10 attention + 30 GDN layers
Hardware	8 $\times$ H20-3e; one book/rank
Software	Python 3.11.13; PyTorch 2.11.0+cu129; Transformers 5.14.1; vLLM 0.26.0+cu129; Triton 3.6.0; CUDA 12.9
KV backend	vLLM fused attention; BF16/Q16; page 128
Document / query	4,095 / 32 tokens; PG-19 train only
Generation	8 output tokens; 39 state-appended input tokens (32 query + 7 generated-token feedback)
Resident counts	$1, 8, 32$; four KV-by-GDN ownership cells
Primary schedule	all requests resident; batch-one sequential round-major calls on one CUDA stream/rank
Validation	96 factor configurations; separate memory/witness cells
Fault campaigns	$N = 2$; the primary all-gates-on campaign has nine live mutants with rebuilt controls; the target-gate-suppression campaign has nine fresh clean plus nine suppressed-gate cases on 8 H20s; the designer–executor-separated campaign has five new clean/mutant pairs on 5 H20s
Historical regression	one previously encountered one-token alias defect; 8 ranks $\times$ 3 fresh $N = 2$ lanes; 3 archived-coordinate reproductions + 5 additional frozen inputs; no mutation

Table 6: Meaning of the live-binding rerun's selected-cell evidence units (archive entry `E-LIVE-BINDING-A`; Appendix J). Counts are per rank unless marked total; all belong to the same formal run.

Unit	Count	Check and authorization
Persistent sources	60	independently enumerated before the actual builder; exact object, storage, descriptor, and content remain fixed
Selected anchors	6/phase	preregistered persistent/request semantic rows; request-owned subset consumes the group-bound lineage capability
Full live/serializer rows	540/phase	$(1 \text{ persistent} + 8 \text{ requests}) \times 60$ rows; live objects are re-enumerated and every serialized field, role, interval, content digest, and phase relation is checked
Builder clone edges	480	8×60 materialized destinations exhaust the captured persistent-rooted ledger and are direct `aten.clone` outputs
Phase artifacts	3	setup, post-transition, and post-generation products are hash-bound, published to stable rank paths, and terminally reread
Clean-memory context separation	12 (96 total)	separate allocator-cell calls are all observed; zero selected-context overlaps mean the selected context did not enter them, not that the wrapper was absent or perturbation-free

With field order (owner, request, layer, family, state), the six anchors are $(\mathrm{Pers}, \varnothing, 0, \mathrm{rec}, 0)$, $(\mathrm{Req}, 0, 0, \mathrm{conv}, 0)$, $(\mathrm{Req}, 0, 1, \mathrm{conv}, 0)$, $(\mathrm{Req}, 0, 2, \mathrm{conv}, 0)$, $(\mathrm{Req}, 0, 2, \mathrm{rec}, 0)$, and $(\mathrm{Req}, 1, 0, \mathrm{rec}, 0)$. The capability covers the five request rows; the persistent row is their source anchor.

Table 7: Cohort-to-claim authorization. Cohorts are never pooled; each row supports only the listed inference.

Cohort	Frozen scope	Authorized claim / explicit exclusion
Primary ownership factorial	8 books; PG-19; full-prefix BF16 KV; no split depth; no quantization; $N = 1, 8, 32$; four KV×GDN cells; 8 decode steps	ownership lifecycle, cross-cell equivalence, selected attention oracle, live-mutant sensitivity, and four-cell allocator endpoints; not the quantized Read path, and not transfer or scheduler evidence
Primary target-5 rerun	same frozen 8-rank, 96-configuration Qwen/H20 factorial; 209,920 attention and 635,520 GDN calls	per-call selected Triton kernel-cache artifact/configuration and eager-route binding at the declared fixed-stack scope; not device-side completion, driver/device, compiled-GDN, ATen/CUDA, performance, or cross-stack evidence
Selected-cell live binding	same frozen 8-rank factorial; stronger hook active only for the $N = 8$ shared-document/materialized-GDN cell; actual builder, production serializer, and direct clone lineage	144 selected rows, 12,960 live-storage rows, 3,840 clone edges, and 24 phase artifacts pass; all 96 clean-memory calls have zero selected-context overlaps; not evidence for other cells or coordinates, malicious runtimes, framework independence, device binaries, or performance
Fresh-process control	one independent $N = 2$ process; one clean path and M1	input reconstruction, cleanup, and one intended-gate reproduction; not pooled with the factorial
Lifecycle transfer	8 books; aligned 4,096-token prefix; $N = 4$; two cancelled slots	exact outputs/state, zero-scrub, exact reuse, and stale-lease rejection on the same adapter; not scheduler, concurrency, or performance evidence
Scheduler interleave	8 books; $N = 4$; page/prefix/tail geometries 64/3,985/17 and 128/4,033/65; cancel slot 3 after round 2	deterministic request-step interleaving, zero-scrub, epoch change, exact reuse, uninterrupted-control equality, and three frozen scheduler-fault gates; not concurrent kernels, continuous-batching performance, detection rate, or production end-to-end correctness

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Table 7 continued.

Cohort	Frozen scope	Authorized claim / explicit exclusion
Independent preproducer slot census	frozen Qwen/H20 geometry and schedule; 180 slots/capture; two $N = 2$ GDN-policy cells and three phases/cell	all 6 captures, 1,080 receiver rows, and 96,660 relations close; omission, duplication, and semantic-relabel controls fail closed; producer manifests/rows are not expected-set authority, but slot-ID-to-live-tensor binding, IPC, and paused capture remain trusted
Dual-producer repeat	same frozen input/protocol and producer implementation; two fresh serial executions and four distinct out-of-process observers	both executions reproduce 1,080 semantic coordinates, byte digests, stable descriptors, and 96,660 relation labels at zero tolerance; not malicious-producer resistance, independent receiver model execution, or cross-stack evidence
Falcon-H1 bounded transfer	`tiiuae/Falcon-H1-0.5B-Base`; Transformers 5.14.1/H20 naive path; $N = 1, 2$; separately executed official Transformers reference; persistent-Q16 and deep-materialized arms	8/8 ranks, 96 generated-token and 96 full-vocabulary FP32-logit comparisons, 13,824 state rows across KV key/value, convolution, and Mamba2 recurrent families, 192 ownership checks, cross-arm/cross-N relations, and registered controls pass; no performance, memory, quality, or other-stack evidence
Bounded two-stream lifecycle	one H20; one 4,033-token document; two 16-token query requests at $N = 2$; two host threads and distinct CUDA streams; pre-cancel and post-reclaim phases; cancellation after synchronization	positive call-interval overlap plus exact four lifecycle outputs, 20 logical-KV layer pairs, two GDN slot aggregates, ownership, scrub, and reuse; not simultaneous kernels, native continuous batching, in-flight cancellation, production scheduling, or performance
GDN transition oracle	1 frozen book/window; global model-layer indices 0, 10, 20, 38; 4 query tokens	clean recurrence outputs/states and four seeded wrong-transition rejections conditional on post-normalization inputs; not all GDN layers or end-to-end semantics
Expanded captured-boundary numerical sweep	2 distinct PG-19 inputs; all 10 attention layers and 12 of 30 GDN layers; 20 attention and 24 GDN clean rows plus one seeded control/row	captured-boundary numerical agreement and rejection of all 44 seeded wrong operators; not upstream-activation, independent-capture, full-model, or end-to-end validation
Target-gate suppression matrix	9 fresh clean + 9 target-suppressed cases; same fixed model/data/$N = 2$; 8 distinct H20s	per-fault comparison with redundant audit gates, exact allowlisted production assertions, and token/FP32-logit equality; M8's injected sentinel is not a detector; not a blind-fault study or rate
Designer–executor-separated faults	separate designer reads only the prior PDF; 5 new frozen $N = 2$ clean/mutant pairs; 5 H20s; all gates enabled	all 5 clean gates pass and all 5 mutants fail first at their frozen primary predicate; per-fault sensitivity only, not a detection rate, natural-defect study, or cross-stack evidence
Historical alias regression	one pre-discovery integration defect; 3 archived coordinates + 5 additional frozen inputs; fresh pre-fix, repaired, and materialized $N = 2$ one-token lanes; 8 H20s	pre-fix binding gate and base violation with output/terminal-state equality, plus storage-clean exact repair; one retrospective defect only, not prevalence, recall, statistical replication, or cross-stack evidence
Mac common-mode control	Apple M4 Pro; six documents/queries; five fresh MLX processes; vanilla dense plus Q-CoMem BF16/Q8/Q4 at depth 7	separates split-family agreement from agreement with vanilla dense; motivation only and separate from the H20 case

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Table 7 continued.

Cohort	Frozen scope	Authorized claim / explicit exclusion
H20 archival Store–F1 validation	60 validation items from Qasper/2WikiMQA (30 each); dense, full-prefix, split Q16, frozen Q4/Q4/Q8, and two mixed policies	raw-first replay of Store, predictions, mean F1, paired bootstrap intervals, and fixed-case failure counts; not source-frozen regeneration, TTFT/TPOT/throughput, Recall, total memory, edge, or broad-benchmark evidence
H20 operating-point run	the same 60 validation items; dense, full-prefix native dtype, full-prefix Q8, and Q-CoMem frozen Q4/Q4/Q8 at $j = 13$ and $j = 7$; one warm-up, three repeats, dataset-length generation, seed 20260903	Store, TTFT, TPOT, mean F1, and paired bootstrap intervals against full-prefix from a separate execution; not pooled with the archival panel, not Recall, total memory, admitted capacity, resident-set, edge, or ForkAudit evidence
H20 deployment timing	source indices 6–9 from each of Qasper/2WikiMQA (8 items total), a subset of the 60-item validation cohort; no-cache dense recompute, full-prefix KV reuse, and Q-CoMem Q8/Q4/mixed variants (six configurations total) with 3 timing repeats	same-stack retained-Store, TTFT, TPOT, tokens/s, and mean F1 comparison from a separate execution; not an independent replication, not pooled with the 60-item panel or primary factorial, and not scheduler/continuous-batching evidence
Matched serving controls	8 validation items from Qasper/2WikiMQA; vLLM 0.26 and SGLang 0.5.17 cache off/on plus 24 fixed-configuration HYPIC timing/quality cells; one H20-3e per item	within-block cache-hit, quality, and client-wall timing context; not pooled with in-process Q-CoMem timings, not a cross-framework leaderboard, and not continuous-batching evidence
HYPIC retained-state receipts	8 Prefix Cache + 8 HYPIC cells on the same fixed TP=1 model/slice; receipt-instrumented SGLang lifecycle	median target-entry-owned physical tensor payload only; not timing, NVML/process memory, capacity, continuous batching, or ForkAudit ownership evidence
Related-work operator transfer	Hydragen at $N = 8/32$ on one captured layer-3 attention family; Palu on one layer with 8 calibration and 1 held-out PG19 row	same-checkpoint numerical and logical-state trade-offs only; not an all-layer implementation, LongBench/quality result, speedup claim, or jointly optimized system
Local replay/storage accounting	one Apple-silicon (M4-class) laptop CPU host; three consecutive fixed-order runs without cache flush; manifest-only core package plus a serial profile with five locally complete supporting verifiers	local CPU replay wall time and logical file/byte counts only; not H20 capture, GPU perturbation, cold-start latency, online inference, compressed-upload size, or an uninstrumented-baseline delta

Table 8: The seven contract targets of Section 4.4: trace coverage, replay verdict, what a pass rules out, and the execution that produced it. A pass is trace-relative and conditional on capture honesty, slot-to-live-tensor binding, and the target boundary of Section 4.4; it is not comprehensive system correctness. Appendix Table 9 gives the decisive witness for each target.

#	Target	Cov.	Verd.	A pass rules out	Execution; scope limit
1	Frozen identity	compl.	pass	a changed source, model, data, query, or protocol between capture and replay	primary rerun; one formal run
2	Prefix immutability	compl.	pass	in-place writes to shared document KV or the GDN base at any phase	primary rerun; one frozen geometry
3	Private ownership	compl.	pass	a request that still writes through an alias of the base or of a peer	primary rerun; live-binding rerun adds six owner-addressed rows per phase in one cell
4	Tail-safe append	compl.	pass	a COW-elided in-place write to the partial final page	primary rerun; one partial-tail geometry
5	Host-side launch/config.	compl.	pass	silent kernel fallback or route contamination on the host side	dispatch rerun; normal launcher return, not device completion
6	Cross-arm equivalence	compl.	pass	an ownership arm that is cheaper because it computes something different	primary rerun; one sequential schedule
7	Cross-N consistency	compl.	pass	a prefix whose reuse depends on how many requests are resident	primary rerun; relational, not an independent oracle

Table 9: Trace coverage and replay verdict for the seven contract targets in fresh frozen primary case. Equality denotes canonical byte-digest identity as recorded. Overall trace coverage is complete and all seven target verdicts pass at their declared fixed-stack scope. A passing primary target verdict remains trace-relative and conditional on capture honesty, slot-ID-to-live-tensor binding, and the target boundary in Section 4.4. The census removes producer rows as expected-set authority only for its bounded GDN cohort; supporting cohorts do not upgrade these targets to comprehensive system correctness.

Target	Trace coverage	Replay verdict	Decisive witness	Boundary
Frozen identity	complete	pass	identical preflight/terminal source ledgers, terminal model rehash, and stable data/query/protocol/execution bindings	one primary-factorial formal run; the dual repeat recaptures only the bounded $N = 2$ GDN cohort
Prefix immutability	complete	pass	physical document-KV digests and GDN base guards across all three phases	one frozen configuration/geometry
Private ownership	complete	pass	all-pairs request–base and request–peer range replay at the primary registered transition; actual-builder, serializer, and direct-clone receipts for the selected cell	pointer-free conservative intervals; stronger live binding covers only six selected rows per phase in one cell
Tail-safe append	complete	pass	one-time 127-token private-tail copy before append	one partial-tail geometry
Host-side launch/config.	complete	pass	per-call ID, selected kernel-cache artifact/configuration, autotuner outcome, GPU/stream, and post-return seal; source-bound GDN eager route	normal launcher return, not device completion; no driver/device, compiled-GDN, or ATen/CUDA attestation
Cross-arm equivalence	complete	pass	tokens, shape/dtype-bound CPU-FP32 logit digests, logical KV, and GDN	one primary sequential schedule
Cross-N consistency	complete	pass	288 all-arm adjacent-fan-out comparisons over 72 rank–query identities	relational, not an independent oracle

Selected numerical oracles, constructed controls, lifecycle/scheduler checks, the independent census and dual-producer repeat, bounded two-stream and Falcon cohorts, and the historical reproduction are supporting checks rather than additional contract targets. Their captured-input, same-adapter, process-separated-but-shared-IPC, second-configuration, and schedule boundaries are reported separately.

E Pointer-Free Storage Witness Schema

Target-to-record dependencies. The following matrix is normative. I, L, E, and F denote the Identity, Lifecycle, Execution, and Target/fault classes in Table 24; parenthesized entries name mandatory fields. Missing any marked field makes that target open. A preregistered geometry may make tail safety not applicable. Target 5 additionally requires an exact per-call ID, selected kernel-cache artifact/configuration, autotuner selection or explicit no-autotuner record, assigned GPU/stream, and post-return seal; missing any applicable field leaves target 5 open. A missing F field invalidates the corresponding fault row, but does not change the separately evaluated seven-target witness.

Enumerated evidence units and call counts. Section 5.5 states the verdicts; the counts behind them are collected here, each tagged with its execution. *Dispatch rerun.* Host-side receipts for 209,920 attention calls record the selected hashed Triton artifact/configuration and the autotune/no-autotuner decision immediately before the original Python launcher, then seal after normal return; another 635,520 receipts record the mutually exclusive GDN eager route and functional cache rebind. The totals factor as $20{,}992 \times 10$ attention calls and $20{,}992 \times 30$ request-GDN calls plus 192×30 one-time prefill calls. They establish host-side selected launch/configuration provenance only, not device-side operator identity, execution, or completion, compiled GDN, underlying ATen/CUDA, or driver/device binaries. *Live-binding rerun.* Table 6 defines each evidence unit. The totals are $8 \times 3 \times 6 = 144$ selected-owner-row checks, $8 \times 3 \times 540 = 12{,}960$ full serializer/live-object rows, $8 \times 480 = 3{,}840$ direct clone edges, and $8 \times 3 = 24$ stable phase artifacts, where the factor 3 is the setup, post-transition, and post-generation *phases* — not the three resident counts that appear in the $8 \times 3 \times 4 = 96$ configuration product of Section 5.5. Here $60 = 30 \times 2$ persistent layer/family coordinates (thirty GDN layers $\times$ recurrent and convolution families) give the $480 = 8 \times 60$ direct clone destinations, and each phase contains the full $540 = (1 + 8) \times 60$ serializer-row universe, one persistent plus eight request owners, with six owner-row anchors. All 96 separately enumerated clean-memory calls occur outside the active selected context; zero selected-context overlaps means only that this context never entered an allocator cell, not that the wrapper was absent or perturbation-free. *Primary rerun.* The 288 adjacent-fan-out comparisons of Section 5.5 span 72 rank–query identities across all arms, and the borrowed-GDN arm's own allocator endpoints are $4.890\,\mathrm{GiB}$ final and $2.229\,\mathrm{GiB}$ shared-document (Table 18). The primary factorial's preregistered candidate-import-free NumPy oracle checks four GDN transitions at global layer indices $0, 10, 20, 38$ against four wrong recurrences (Table 28), and its supporting cohorts derive 1,080 GDN observations and 96,660 relations from a source-distinct expected-slot census (Appendix H).

Target	I	L	E	F	Blind predicate
Frozen identity	code, model, data, protocol; role map	—	—	N/A	exact bytes and complete roles
Prefix immutability	document/base roles; start digests	setup and final guards	canonical content digests	N/A	start = final
Private ownership	request/base/private roles	setup, registered-transition, final bindings	normalized storage ranges	N/A	non-vacuous all-pairs disjointness
Tail-safe append	page geometry; document/private roles	first append event	ordered append and copied-tail receipt	N/A	one private copy before write
Host-side launch/config.	callable, source, protocol, selected artifact/config.	post-return seal; GPU/stream assignment	ordered call, query, mask, position, append, scale, autotuner outcome, eager route	N/A	exact per-call binding; no fallback or route contamination
Cross-arm equivalence	factor, query, and schedule identities	final bindings	token, logit, logical-KV, GDN records	N/A	canonical semantic equality
Cross-N consistency	query and fan-out identities	—	the same canonical semantic records	N/A	smallest-N equals each larger N
Fault sensitivity (supporting)	frozen target/gate map	matched-control rebuild and cleanup	injected outcome and failed predicates	required	target, injection, expected gate, outcome

Each tensor-view row has the exact fields `storage_id`, `shape`, `stride`, `storage_offset`, `dtype`, `device`, `storage_nbytes`, `tensor_nbytes`, and `content_sha256`; absolute addresses are forbidden. Within each phase, the producer assigns `storage-0000`, `storage-0001`, and so on in first-observation order to backing storages. Reusing an ID with a different device or storage size is invalid. Phase capture IDs are unique, whereas request and persistent lifecycle-guard IDs must remain stable across setup, the registered transition, and final generation.

Let element size be e, storage offset o, shape s_i, and stride t_i. The independent replay initializes $m = M = o$ and, for every dimension, computes

$$d_i = (s_i - 1)t_i, \qquad m \mathrel{+}= \min(d_i, 0), \qquad M \mathrel{+}= \max(d_i, 0). \tag{5}$$

The conservative half-open byte interval is

$$I = [em,\, e(M+1)), \tag{6}$$

which must be non-negative and contained in `storage_nbytes`; also `tensor_nbytes`$= e\prod_i s_i$. Two rows overlap iff their normalized storage IDs match and $I_a^{\text{start}} < I_b^{\text{end}}$ and $I_b^{\text{start}} < I_a^{\text{end}}$. Exact aliasing additionally requires equality of interval, geometry, dtype/device, storage and tensor sizes, and content digest. For correctly live-bound rows at one capture, if represented views sharing physical backing receive the same normalized ID and their shape, stride, offset, and element size exactly describe affine-strided views, I contains every touched byte. The overlap test then cannot miss a shared byte between represented rows, although interleaving may cause false positives. Omitted rows, between-capture changes, and wrapper or storage-ID violations remain outside the claim.

At setup, every recurrent-state coordinate must either exactly alias the persistent state (borrowed-base arm) or be disjoint from it (materialized arm). At the primary registered 32-token transition boundary, a completed request's 60 mutable rows must be disjoint from the base and every peer; incomplete borrowed requests must remain exact aliases. At final, all request-local GDN states must be disjoint from the persistent GDN base and every peer; document KV remains intentionally shared. Independent binding-token receipts additionally require unchanged tokens for incomplete requests and changed binding/storage tokens for completed requests. The package applies this schema to all 96 timelines and 288 phase artifacts; nine adversarial tests cover exact aliasing, partial overlap, adjacent intervals, negative strides, conflicting ID reuse, pointer injection, capture/guard reuse, and cross-coordinate request–base and request–peer aliasing.

F Executed Fault Rows and Invariant Map

The row-level campaign is kept out of the main paper because it supports diagnosis and audit coverage rather than a separate headline result. Separately, the CPU governance suite passes 17 tests and rejects 28 explicit serialized-artifact tamper cases plus its registered helper/launcher faults. That result concerns schema and orchestration fail-closure only; it is not GPU semantic or performance evidence.

Table 10: Separate primary all-gates-on reference. Every separately rebuilt matched-clean cell passed; every mutant reached its target gate fixed before execution and was restored before disposal.

ID	Injected live fault	Expected/observed gate	Primary outcome
M1	reservation alias	`KV_RESERVATION_DISJOINT`	target gate reached
M2	sequence swap	`KV_SEQUENCE_ID`	target gate reached
M3	omit tail COW	`KV_TAIL_COW`	target gate reached
M4	GDN–base alias	`gdn_completed_vs_base_disjoint`	target gate reached
M5	GDN peer alias	`gdn_completed_vs_peers_disjoint`	target gate reached
M6	position off-by-one	`POSITION_CANONICAL_VALUES`	target gate reached
M7	materialized mask	`MASK_CONTRACT`	target gate reached
M8	wrong callable	`KERNEL_CALLABLE_ID`	target gate reached
M9	dense KV view	`KV_PAGED_VIEW`	target gate reached

Table 11: Same-system outcomes for nine designed faults. The primary campaign enables all gates; a separate target-gate-suppression campaign suppresses only the named gate. "Same" is exact and "N/O" an earlier stop. Rows are not a detection rate; M8's injected sentinel is not a production detector.

Fault / target contract	All gates on	With target gate suppressed	Token / FP32 logits
M1 Reservation ownership	target gate	completes	Same / Same
M2 Sequence binding	target gate	completes	Same / Same
M3 Tail COW	target gate	other audit: active-block ownership	N/O / N/O
M4 GDN base isolation	target gate	other audit: peer isolation	N/O / N/O
M5 GDN peer isolation	target gate	completes; logits change ($L_\infty = 0.75$, rel. $L_2 = 0.0314$)	Same / Changed
M6 Positions	target gate	completes	Same / Same
M7 Mask contract	target gate	completes	Same / Same
M8 Callable identity	target gate	injected payload sentinel	N/O / N/O
M9 Paged-KV view	target gate	paired-view production assertion	N/O / N/O

Designer–executor-separated fault detail. The five fixed per-fault outcomes summarized in the main text are shown below; they remain constructed controls rather than a defect-population sample.

Table 12: Designer–executor-separated constructed fault outcomes. A separate designer saw only a frozen earlier manuscript PDF and fixed all five definitions, loci, payloads, clean requirements, and expected first predicates before executor preparation or candidate output. The cases are separated from executor preparation and outcome inspection, not from the predicate vocabulary. All matched clean cases pass; every mutant completes its registered horizon and is rejected first by the frozen primary predicate. Semantic comparisons are secondary: "Same" means canonical exact equality with the matched clean case, KV/GDN denote terminal logical state, and "N/C" means full-logit call cardinality is not comparable. These five fixed outcomes are not a population detection rate.

Frozen fault	Isolated mutation	First failed predicate	Tokens / logits / KV / GDN
HF01 delayed tail detach	append write precedes the otherwise present tail copy	tail-copy-before-first-write	Same / Same / Same / Same
HF02 inactive-lane scribble	one bit in the unused lane of a shared physical K page	physical-prefix immutability	Same / Same / Same / Same
HF03 duplicate commit	one correct feedback call commits twice; first output discarded	ordered-call cardinality	Same / N/C / Changed / Changed
HF04 effective-scale drift	first registered attention scale is exactly doubled	attention effective scale	Changed / Changed / Changed / Changed
HF05 stale GDN binding token	storage correctly rebinds; lifecycle token remains stale	completed binding-token advance	Same / Same / Same / Same

Table 13: Representative failure families and intended checks. Rows naming M1–M9 refer to the nine executed primary-factorial live mutants; the other rows summarize schema, relational, numerical, or replay checks. None is a complete oracle.

Check family	Intended sensitivity and observed boundary
Frozen identity	wrong model/data/query bank, changed protocol, stale shard (schema adversaries)
Prefix immutability	M3 omits partial-tail COW; physical document digest and ownership gates fail
Private ownership	M1 reservation alias, M4 request–base GDN alias, M5 request–peer GDN alias
Sequence/view binding	M2 sequence swap and M9 dense-KV replacement
Host-side selected launch/configuration	M6 position offset, M7 materialized mask, M8 wrong callable
Factorial exactness	behavior changes when either KV or GDN setup ownership changes
Bounded numerical oracle	numerical error in a selected fused full-attention result
Raw replay	internally inconsistent trajectory, storage, allocator, or aggregate fields

The primary all-gates-on execution is a first-gate localization study. Every registered gate fires before the injected path produces a post-injection token, logit, logical-KV digest, or GDN digest; its output cells are therefore N/O rather than misses. Lifecycle/ownership/storage receipts supply the first gate for M1 and M3–M5; request-binding/call receipts supply it for M2 and M6–M9. The full audit rejects all nine at their predeclared gates under passing matched controls. The separate target-gate-suppression matrix then suppresses one named gate per mutant while leaving the remaining observers active. It provides the per-fault comparison in Table 11, but does not estimate a population rate, unseen-fault coverage, or false-positive rate.

G Relation to Prior Work and Audit Obligations

Table 14: Primary-source positioning. "Evidence focus" describes what the cited work evaluates; it is not a claim that the work lacks all other checks.

Work	Primary mechanism	Evidence focus / relation to this work
PagedAttention	Paged KV allocation and sharing (Kwon et al., 2023)	Serving throughput and memory utilization; supplies the block-sharing substrate.
SGLang / Prompt Cache	Prefix reuse in structured or modular prompts (Zheng et al., 2024; Gim et al., 2024)	Runtime performance and output behavior; establish reuse and positional handling as prior art.
CoMem	One persistent residual per token at split j; bounded selection; suffix replay (Liu et al., 2026a;b)	Supplies the split-replay foundation; we do not claim that primitive as new.
Q-CoMem (ours)	Joint low-bit split residual and required lower KV/convolution/recurrent state	Adds complete hybrid-state packing, a capacity–latency–quality evaluation, and an auditable request-local ownership boundary; not a model-weight compressor or production scheduler.
Marconi	Admission and eviction for hybrid-model prefixes (Pan et al., 2025)	Hybrid cache hit rate and time to first token; closest state-heterogeneous caching system, but a different policy-level contribution.
SCBench	Lifecycle-oriented KV-cache benchmark (Li et al., 2025b)	Breadth over tasks and cache phases; exposes breadth absent from our bounded declared configurations.
MT-DLComp / CRADLE	Metamorphic compiler testing and cross-backend differential checking (Xiao et al., 2022; Pham et al., 2019)	Direct testing precedents; our audit adds ownership witnesses and a layerwise dense oracle, but not an independent end-to-end backend.
DNNMem	Component-wise GPU-memory accounting (Gao et al., 2020)	Supports denominator separation; it is an estimator, not a request-ownership audit.
ForkAudit	Typed, phase-indexed ownership/call/accounting trace with explicit coverage semantics	Validates the ownership component through a mutation-tested KV-by-GDN factorial, bounded host-side call receipts, a preproducer expected-slot census, numerical sweeps, and one bounded Falcon-H1 comparison; it is not a security attestation or general end-to-end correctness oracle.

Table 15: Closest established precedent for each audit obligation. The final column states the narrower integration contributed by ForkAudit; it does not claim that any individual mechanism is new.

Audit obligation	Closest established precedent	Integrated obligation in this work
Frozen identity	Controlled variants in metamorphic and differential testing (Chen et al., 2018; Xiao et al., 2022; Pham et al., 2019)	Bind model, data, query bank, callable contract, and protocol bytes to every ownership receipt.
Document immutability and tail COW	PagedAttention block sharing and copy-on-write for shared prefixes (Kwon et al., 2023)	Replay physical document pages, padding, active tables, and the partial-tail detach event.
Recurrent-state ownership	Hybrid recurrent states and their in-place update constraint (Yang et al., 2025; Pan et al., 2025)	Witness request–base and request–peer ranges before and after the frozen schedule's registered transition boundary.
Host-side selected launch/configuration	Cross-backend differential checking exposes implementation-dependent behavior (Pham et al., 2019)	Hold the backend fixed; record callable, positions, mask, scale, and append history; and bind the selected Triton kernel-cache artifact/configuration per attention call while keeping GDN at an explicit eager-route boundary.
Cross-arm and cross-N relations	Metamorphic relations over semantics-preserving transformations (Chen et al., 2018; Xiao et al., 2022)	Treat ownership policy and resident fan-out as transformations that must preserve registered request observables.
Bounded numerical oracle	Independent backend comparison (Pham et al., 2019)	Reconstruct selected attention inputs for dense IEEE-FP32 math and selected post-normalization GDN inputs for a candidate-import-free NumPy recurrence.
Memory denominators	Component-wise GPU-memory accounting (Gao et al., 2020)	Keep page geometry, allocator current/peak, and unmeasured process/service capacity as separate claim domains.

H SUPPORTING CONTROL EXPERIMENTS

Mac common-mode motivation. The archival Apple-M4-Pro control is kept unpooled from the H20 experiments and serves only to show why split-family agreement does not replace a dense or independently implemented oracle.

Table 16: Unpooled Mac motivation context (Apple M4 Pro, five fresh MLX processes; median over 30 query executions at split depth 7). Vanilla dense is the external reference. Q-COMEM BF16/Q8/Q4 variants use the same selected documents. Store size is the six-document residual corpus; online time includes query Write, dequantization, and suffix Read. This small teacher-forced demonstration set is not primary ForkAudit evidence and supports no cross-hardware or production-speed claim.

Method	Store (MiB)	Online (ms)	Speedup	Top-1 vs. dense
Vanilla dense prompt	–	194.70	1.00×	100.00%
Q-COMEM BF16 split	1.998	168.88	1.15×	71.89%
Q-COMEM Q8 split	1.061	169.17	1.15×	71.89%
Q-COMEM Q4 split	0.562	169.25	1.15×	71.89%

Q8/Q4 retain 100% top-1 agreement with the BF16 split path, yet all split rows inherit only 71.89% agreement with vanilla dense. The speed column is a small teacher-forced mechanism benchmark, not production RAG throughput.

Historical alias regression and repair. The pre-fix, repaired, and materialized lanes isolate one previously observed recurrent-state alias. This is a fixed mechanism study, not a natural-bug frequency or detector-recall estimate.

Table 17: Retrospective reproduction and repair of one historical alias regression. Ranks 0–2 reuse three archived input coordinates; ranks 3–7 are additional frozen inputs, not additional defects or statistical replicates. Semantic columns compare each path with a fresh materialized control. "Base inv." is within-path persistent-base immutability, retained as a conventional state-invariant catch.

Coordinates	Path	Ownership result	Token	FP32 logits	Request GDN	Logical KV	Base inv.
Archived (3)	Pre-fix	binding gate 3/3	exact 3/3	exact 3/3	exact 3/3	exact 3/3	fail 3/3
Archived (3)	Repaired	storage-clean 3/3	exact 3/3	exact 3/3	exact 3/3	exact 3/3	pass 3/3
Additional (5)	Pre-fix	binding gate 5/5	exact 5/5	exact 5/5	exact 5/5	exact 5/5	fail 5/5
Additional (5)	Repaired	storage-clean 5/5	exact 5/5	exact 5/5	exact 5/5	exact 5/5	pass 5/5

Primary allocator endpoints. The four ownership cells use one frozen post-priming allocator denominator and are kept separate from deployment Store.

Table 18: Primary 8×H20-3e allocator results at $N = 32$ (median across ranks; rank values coincide). Full-copy is the factorial control; the paged-prefix row is the closest same-stack prefix-sharing baseline. The remaining rows isolate recurrent-base ownership. All cells retain the persistent document source. Values are GiB above the frozen post-priming baseline; generation increment is the additional peak above allocation present at generation start.

Empirical arm	KV / GDN setup	Final	Peak	Gen. incr.
Full-copy factorial control	Full-copy KV / Materialized GDN base	4.901	4.920	0.019
GDN ownership ablation	Full-copy KV / Borrowed GDN base	4.890	4.907	1.951
Paged-prefix baseline	Shared-doc KV / Materialized GDN base	2.229	2.843	0.019
Audited hybrid fork	Shared-doc KV / Borrowed GDN base	2.229	2.843	1.950

Published systems context. Table 19 reports each cited system only in its native protocol and keeps all non-commensurate metrics outside our matched panels. This unpooled table is literature context, not ForkAudit evidence.

Table 19: Unpooled published-systems context in each paper's native protocol. These values show the scale and evaluation breadth of related systems, *not* a cross-paper leaderboard or ForkAudit evidence: models, hardware, batching, baselines, and metrics differ. Consequently, no row is merged with our same-stack H20 Table 21, and the absence of a common score must not be read as a negative result.

System	Native method and study setting	Reported efficiency in its native protocol	Quality and comparability boundary
PagedAttention / vLLM (Kwon et al., 2023)	Paged KV allocation plus serving engine *Setting:* LLaMA-7B on A10G and LLaMA-13B on A100-40GB; vLLM serving engine	2–4× serving throughput at similar latency vs. FasterTransformer/Orca	No common LongBench F1; exact cache-management method
Prompt Cache (Gim et al., 2024)	Modular attention-state reuse *Setting:* Llama2/MPT/Falcon; RTX 4090, A40, and A100; HuggingFace-style prototype	1.5–3.1× TTFT (CPU-resident modules); 3.7–11.7× (GPU-resident) on its 12-dataset LongBench suite	LongBench accuracy evaluated, but with different models and hardware
SGLang / RadixAttention (Zheng et al., 2024)	Prefix reuse plus structured-program runtime *Setting:* Llama-7B/Mixtral-8x7B and structured programs; A10G/A100; SGLang	Up to 6.4× throughput vs. prior inference systems	Application-suite result; no common LongBench F1
ChunkAttention (Ye et al., 2024)	Prefix-aware KV chunks and attention kernel *Setting:* attention microbenchmarks; mainly A100-80GB, plus A100-40GB/RTX 4090; CUDA 11.8	3.2–4.8× attention-kernel speedup for 1K–4K shared prompts	Kernel result; no end-to-end common task score
Hydragen (Juravsky et al., 2024)	Exact batched shared-prefix attention *Setting:* CodeLlama-13B/34B shared-prefix batches; A100 (with H100/L40S microbenchmarks); custom HuggingFace runtime	Up to 32× CodeLlama-13B end-to-end throughput in large shared-prefix batches	Exact attention construction; different model and batching regime
Palu (Chang et al., 2025)	Low-rank latent KV with reconstruction/fusion *Setting:* Llama/Mistral/LongChat quality studies; Llama-2-7B latency on one RTX 4090; HuggingFace plus custom kernels	50% KV compression and up to 1.89× RoPE-attention speedup; up to 2.91× with quantization	Llama/Mistral/LongChat native study; our bounded Qwen3.5 operator transfer is reported separately
Preble (Srivatsa et al., 2025)	Distributed prompt-sharing scheduling *Setting:* Mistral-7B/Llama-3-70B request traces; two–eight GPUs across A6000/H100 clusters; distributed serving	1.5–14.5× average-latency and 2–10× p99-latency improvement	Distributed serving traces; no common task score
Marconi (Pan et al., 2025)	Admission/eviction for hybrid-model prefixes *Setting:* Attention–Mamba2 7B geometry; LMSys/ShareGPT/SWE-bench traces; CloudLab policy simulation	Up to 34.4× token-hit rate and up to 71.1% or 617 ms lower TTFT	Hybrid models and trace-driven policy study; no common LongBench F1
KVTuner (Li et al., 2025a)	Offline sensitivity-aware search over layer-wise mixed-precision KV bit pairs *Setting:* Llama-3.1-8B-Instruct and Qwen2.5-7B-Instruct; mathematical- and general-reasoning suites	Nearly lossless 3.25-bit mixed-precision KV for Llama-3.1-8B-Instruct and 4.0-bit for Qwen2.5-7B-Instruct; up to 21.25% higher maximum inference throughput vs. KIVI-KV8	Layer-wise bit search for a dense KV cache; it does not quantize a split residual or recurrent state, and no common Qwen3.5 slice exists
HYPIC (Liu et al., 2026c)	Position-independent caching for hybrid full/linear-attention models *Setting:* Qwen3.5-35B-A3B main panels at TP=2 on 8 H20-3e; SGLang 0.5.14	3.25× lower TTFT on average and 1.66× higher QPS over Prefix Cache	Four models/five workloads; our same-slice TP=1 timing and retained-state controls are reported separately

Same-checkpoint transfer detail. Table 20 is the matched complement to the published context table: it uses our frozen Qwen3.5 checkpoint and H20-3e, binds the official Hydragen and Palu source revisions, and independently replays all raw operator sidecars. Because it does not execute all layers or LongBench, it is kept separate from Table 21.

Table 20: Bounded same-checkpoint H20 transfers. Hydragen uses frozen Qwen3.5 layer-3 post-RoPE attention; Palu uses eight 256-token PG19 calibration rows and one disjoint held-out row. Errors are relative ℓ_2 against the matched uncompressed operator; state ratios are logical selected-operator ratios, not process memory or end-to-end results.

Port	Setting	Matched numerical result	Logical state result	Timing / boundary
Hydragen port	$N = 8$	Hydragen–dense rel. ℓ_2 = 0.00255	8.50 vs. 64.48 MiB (7.59× smaller)	0.287 vs. 0.0965 ms; no speedup
Hydragen port	$N = 32$	Hydragen–dense rel. ℓ_2 = 0.00262	10.00 vs. 257.94 MiB (25.80× smaller)	0.290 vs. 0.259 ms; no speedup
Palu port	rank 64/head	held-out K/V $0.487/0.549$ (plain $0.711/0.839$)	512 vs. 2,048 B/token (4.00× smaller)	projection only; no fused kernel
Palu port	rank 128/head	held-out K/V $0.317/0.364$ (plain $0.543/0.668$)	1,024 vs. 2,048 B/token (2.00× smaller)	projection only; no fused kernel
Palu port	rank 192/head	held-out K/V $0.149/0.194$ (plain $0.341/0.459$)	1,536 vs. 2,048 B/token (1.33× smaller)	projection only; no fused kernel

Hydragen avoids prefix replication but has no speedup here. Palu improves over plain SVD, but residual error forbids an end-to-end quality claim. Neither is a jointly optimized combined system.

H.1 Extended Deployment and Serving Context

Table 2 isolates the matched Transformers block used for the main Q-CoMem claim. The expanded table below repeats that block only to place separately measured SGLang/HYPIC context beside it. The blocks use different runtimes and harnesses, so timing and F1 are interpreted within block and are not pooled. These serving controls are not ForkAudit validation evidence and do not establish robust quality preservation.

Table 21: Illustrative same-checkpoint H20 retained Store and mean-F1 point estimates on an eight-item Qasper/2WikiMQA slice. All rows use Qwen3.5-35B-A3B, one H20-3e per item, a 4,096-token cap, greedy decoding, and at most 32 generated tokens. The Transformers (three repeats/item) and HYPIC (Liu et al., 2026c) blocks are unpooled; HYPIC timing/F1 and Store use separate 24- and 16-cell TP=1 cohorts. Timing is compared only within blocks; no robust-quality, capacity, or ForkAudit-ownership claim is made.

Method	Persistent state	Store (MiB)	TTFT (s)	TPOT (ms)	tok/s	F1
Transformers 5.14.1 in-process deployment cohort						
No-cache dense recompute	–	–	0.649	648.75	1.36	39.42
Full-prefix KV cache	Q16	140.34	0.163	54.11	13.38	39.14
Q-CoMem state	Q8	**15.89**	0.673	55.44	7.15	39.14
Q-CoMem mixed state	Q4/Q4/Q8	10.01	0.674	54.97	7.19	39.01
Q-CoMem per-layer mixed	mixed	**9.74**	0.673	55.07	7.19	39.11
Q-CoMem state	Q4	8.41	0.674	55.02	7.19	36.09
HYPIC commit 98147c0 timing/F1; receipt-instrumented Store; SGLang 0.5.14, TP=1						
Official HYPIC code: full recompute	–	–	0.217	49.21	14.77	39.14
Official HYPIC code: prefix cache	Radix prefix	139.53	0.072	50.15	19.07	39.14
HYPIC (transition + seam 8)	segment transition	324.09	0.101	52.13	17.67	39.63

Bold marks the two illustrative Store–F1 points; these are not a headline validation result. Relative to full-prefix Q16, Q8/per-layer mixed reduce Store $88.68\%/93.06\%$; unrounded mean-F1 deltas are $0.000/{-0.022}$ (39.115 vs. 39.137 for mixed). "–" means no retained entry; Q16 is the unpacked reference, BF16 for the residual and attention KV and the stack's native FP32 for GDN recurrent state. Full-prefix Q16 is Q-CoMem's matched semantic comparator; the archived rows show no-cache recompute sharing the same document/query boundary but emitting different greedy token sequences on a few items. Q4/Q4/Q8 gives residual/attention/linear bits. The per-layer policy (Q4 residual, layer bits $[8, 8, 4, 4, 8, 8, 8]$) was selected on a separate, disjoint four-item calibration set and frozen before this eight-item validation. Store is median retained-document tensor payload and excludes metadata, pools, process deltas, and capacity. Published HYPIC panels use TP=2; this adaptation uses TP=1.

Matched serving-framework and HYPIC context. Table 22 gives separately timed vLLM and SGLang cache-off/on controls plus the official-code HYPIC Full Recompute, Prefix Cache, and full HYPIC arms under the shared Qwen3.5 streaming boundary. It remains a timing/quality-only panel and authorizes only within-framework interpretation; the separate 16-cell retained-state receipt cohort appears in Table 21. vLLM and SGLang each record 8/8 prefix hits and preserve 8/8 cache-off predictions. Within its TP=1 block, HYPIC gives 39.63, 0.101 s, and 17.67 tokens/s; published Qwen3.5 panels use TP=2, so this is an adaptation. Its 24 timing cells bind a 4,404-entry terminal artifact ledger. Across the 16 retained-state cells, Prefix Cache retains a median 139.53 MiB and HYPIC retains 324.09 MiB, against 15.89 MiB for Q-CoMem Q8 under the same byte-range-union estimand. We report the three levels rather than their ratios: the SGLang payload is quantized to 128-token pages (across the eight documents it takes only three distinct values, all pairwise gaps integer multiples of the 1,310,720-byte ten-layer page) whereas the in-process payload is element-exact and varies continuously with document length. The one object both stacks hold — the retained full prefix for the same checkpoint, slice and cap — reads 139.53 MiB under the SGLang receipts and 140.34 MiB in-process, a 0.58% difference, which bounds the cross-runtime inclusion-rule discrepancy at that single design point. This remains single-stream; it does not measure continuous batching.

Table 22: Unpooled timing/quality-only serving context on Qwen3.5 and the same eight-item Qasper/2WikiMQA slice. Each row uses one H20-3e per item, a 4,096-token cap, greedy decoding, and at most 32 generated tokens. Times are OpenAI-compatible streaming client wall-clock; F1 is the eight-item mean. Hit is interpreted only against the disabled/full-recompute reference inside each framework block. No cross-framework speedup, retained-state, scheduler-throughput, capacity, or broad-quality claim is made.

Framework	Prefix reuse	F1	TTFT (s)	TPOT (ms)	tok/s	Hit
vLLM 0.26	off	39.63	1.445	56.31	4.70	–
vLLM 0.26	align on	39.63	0.179	53.99	14.00	8/8
SGLang 0.5.17	off	39.14	0.283	53.38	12.73	–
SGLang 0.5.17	Radix on	39.14	0.146	53.15	15.83	8/8
HYPIC code	full recompute	39.14	0.217	49.21	14.77	–
HYPIC code	prefix cache	39.14	0.072	50.15	19.07	8/8
HYPIC	transition+seam 8	39.63	0.101	52.13	17.67	8/8

vLLM uses experimental Mamba `align`; SGLang 0.5.17 uses `extra_buffer` Radix caching. The HYPIC rows use the authors' commit 98147c0 on SGLang 0.5.14, TP=1, full `transition_rope_recompute`, and an eight-token seam. Its full and prefix rows are controls from that same codebase; published TP=2 results and the separate retained-state receipt cohort are not inserted here.

Hybrid-prefix policy context. The separate preregistered Marconi trace extends the eight frozen workloads with a fixed synthetic follow-up turn and evaluates only prefix-policy token reuse under the artifact's native Attention–Mamba2 geometry. It neither reruns LongBench quality nor supplies Qwen3.5 serving speed, so it is not pooled with the preceding panel or the primary audit.

Table 23: Unpooled preregistered policy-simulator context on a 128-event synthetic multi-turn extension of the eight frozen workloads. Values are token-hit percentages under the official Marconi artifact's native Attention–Mamba2 geometry. The wrapper calls each policy's own eviction routine after insertion to close actual tree size to the declared budget. These are not Qwen3.5 quality, latency, or throughput measurements and do not validate the primary ownership audit.

Policy	5 GB	10 GB	20 GB
vLLM+	2.07	6.18	46.83
SGLang+	90.78	93.86	93.86
Marconi	**93.86**	93.86	93.86

Table 24: Normative record classes. "Mandatory" means that absence makes the corresponding target open rather than silently not applicable.

Record class	Mandatory payload	Independent replay
Identity	frozen code, model, data, protocol, and object-to-role map	exact bytes and role coverage
Lifecycle	setup, registered-transition, and final inventories and bindings	normalized ranges, aliases, and disjointness
Execution	ordered call/append receipts, semantic records, named allocator snapshots	frozen schedule, digests, and phase equations
Target/fault	predeclared target, gate, matched control, injection, and outcome	failed-predicate set and classified outcome

Aligned cancellation and reclamation lifecycle. The separate lifecycle-transfer cohort uses eight ranks, eight distinct PG-19 training books, an aligned 4,096-token prefix, and $N = 4$. It is not pooled with the primary factorial.

Table 25: Lifecycle-transfer obligations on the existing Qwen3.5/vLLM Q16 adapter. Counts are deterministic rank outcomes, not stochastic replicates.

Obligation	Result
Frozen static/code/model/weight/raw bindings	pass; 14/14 weights, 8/8 shards
Full-vocabulary logits and generated tokens vs. uninterrupted control	exact, 8/8
Final logical KV and final GDN state	exact, 8/8
Aligned prefix, immutable document, and zero partial-tail copy	pass, 8/8
Cancelled-slot zero-scrub and exact reservation reassignment	pass, 8/8
Stale epoch-0 rejection and matched epoch-1 acceptance	8/8; 0 escapes, 0 wrong gates

Scheduler-managed request-step interleaving. The separate formal extension keeps four requests resident, cancels slot 3 after its second round, zero-scrubs and increments its lease epoch, reassigns the exact reservation, and completes the replacement interleaved with the three survivors. It is not pooled with the primary factorial or the aligned lifecycle-transfer cohort.

Table 26: Formal scheduler-managed request-step interleaving on the frozen Qwen3.5/ForkAudit vLLM-Q16 stack. Clean counts are deterministic rank outcomes; fault counts are preregistered expected-gate outcomes, not a detection rate.

Case	Frozen geometry or intervention	Outcome
Clean geometry 1	page 64; 3,985-token prefix; 17-token document tail; 18-event cancel/replacement lifecycle	all semantic/storage gates and uninterrupted-control comparisons pass, 8/8
Clean geometry 2	page 128; 4,033-token prefix; 65-token document tail; same lifecycle	all semantic/storage gates and uninterrupted-control comparisons pass, 8/8
FH1	dispatch cancelled epoch-0 lease	`STALE_SLOT_LEASE`, 16/16
FH2	dispatch lease under another request	`LEASE_REQUEST_MISMATCH`, 16/16
FH3	reclaim before zero scrub	`RECLAIM_NOT_ZERO`, 16/16

Candidate-import-free GDN transition oracle. The oracle uses one frozen PG-19 window and four query-time GDN transitions at global model-layer indices 0, 10, 20, and 38. A single preregistration frozen before the successful run binds the post-native-q/k-normalization boundary, scale $1/\sqrt{128}$, tolerances, and seeded faults. Its NumPy reference starts at that captured boundary; no candidate module is imported by the reference.

Table 27: PDF-visible oracle selection and replay boundary. Every listed row was locked before its successful candidate execution; neither oracle validates upstream activation construction or end-to-end model behavior.

Oracle	Frozen rows / selection rule	Captured boundary	Decision rule	Replay availability
Attention	rank r uses layer $4r + 3$ and round r, for $r \in \{0, \ldots, 7\}$; request 0, $N = 1$, shared-document KV + borrowed GDN; frozen rank/layer/round map	post-RoPE Q/K/V, positions, mask, scale, and ordered append shadows	dense CPU-FP32 rel. $L_2 \leq 0.005$	8 raw rows and numerical sidecars; primary-package one-command replay
GDN	global model layers $0, 10, 20, 38$; four query tokens from one frozen PG-19 window	post-native-q/k-normalization recurrence; scale $1/\sqrt{128}$	output rel. $L_2 \leq 0.005$; state rel. $L_2 \leq 10^{-4}$; seeded fault must reject	4 clean + 4 fault decisions from 40 sidecars; separate one-command replay

Table 28: Bounded GDN recurrence oracle. Relative errors compare native outputs and final states with the candidate-import-free NumPy FP32 recurrence.

Global layer	Output rel. L_2	State rel. L_2	Clean	Seeded fault	Rejected
0	0.001584	1.23×10^{-7}	yes	omit decay	yes
10	0.001652	1.36×10^{-7}	yes	complement β	yes
20	0.001572	1.39×10^{-7}	yes	pre-decay memory	yes
38	0.001638	1.30×10^{-7}	yes	roll value heads	yes

Table 29: Preregistered expanded captured-boundary numerical sweep. Clean rows use candidate-import-free CPU/NumPy FP32 replay; every clean row has one seeded wrong-operator control. Attention begins after RoPE and GDN begins after native q/k normalization, so the table does not validate upstream activation construction, independent capture, or end-to-end model behavior.

Operator	Inputs	Layers	Clean rows	Positions / transitions	Max output rel. L_2	Controls rejected
Attention	2	10/10	20/20	160 positions	0.001897	20/20
GDN	2	12/30	24/24	192 transitions	0.002073 (2.08×10^{-7} state)	24/24

Full target-gate-suppression matrix. The preregistered target-gate-suppression execution contains all M1–M9 faults and a fresh matched clean path for each. Across eight distinct H20s, all 18 cases complete their declared horizon or an allowed classified stop, all 24 FP32 sidecars verify, and no case is operationally invalid. Five mutant paths complete semantics: token equality catches 0/5, exact logits catch M5 only ($L_\infty = 0.75$, relative $L_2 = 0.0314$), and M1/M2/M6/M7 leave both exact. M3/M4 reach other audit gates; M9 reaches the paired-view production assertion; M8 reaches the exact injected payload sentinel, which is not a production detector. The frozen aggregator initially read the inherited execution identifier from a detached-manifest field that the manifest schema omits. A disclosed postexecution wrapper derives that identifier from the canonical receipt embedded in all eight manifest-bound primary-package shards, changes only the generated comparison-row `run_id` from null to the preregistered value, and reproduces the summary byte-for-byte without modifying candidate outputs or frozen sources.

I Expanded Limitations and Claim Boundary

The primary factorial covers one model, one H20 hardware family, one page size, 16-bit KV, and $N \leq 32$. Its ownership conclusion is tied to one 4,095-token partial-tail geometry, a registered 32-token continuation block, and the subsequent frozen sequential schedule; it is not a claim about an arbitrary first token or transition length. The lifecycle cohort adds one aligned 4,096-token geometry and a fixed cancel/scrub/reclaim sequence. The scheduler-interleave extension adds two geometries and deterministic request-step interleaving, but those requests execute sequentially on one CUDA stream.

The separate two-stream cohort has two host workers and distinct CUDA streams with positive full-model-call interval overlap. Its intervals include queuing, resource waits, and host-enqueue gaps; there is no per-kernel overlap trace. Cancellation occurs only after synchronization, so the cohort does not test arbitrary or in-flight reclamation. The original out-of-process GDN observer removes same-process reconstruction and live judgment labels. The later preproducer census independently fixes the expected slot set, but correct slot-ID-to-live-tensor binding remains trusted; both sides also trust PyTorch tensor/storage and CUDA-IPC semantics, and mutation is paused for each snapshot. The separate selected-cell hook binds an actual builder return through the production serializer and direct clone lineage, but still runs with the producer in one honest process. These cohorts do not supply OS/driver allocation ground truth, independent receiver-side model execution, KV/kernel attestation, or protection against a transient write followed by restoration.

No cohort evaluates native continuous/ragged batching, eviction, multi-document serving, general scheduler cancellation, process-level NVML memory, OOM capacity, or a broad downstream benchmark. Full-copy KV and materialized GDN bases are controlled ownership factors rather than optimized production baselines. The GDN path is the Transformers Torch implementation, not an optimized recurrent kernel. The target-5 rerun closes only the declared honest-process host-side boundary: attention calls bind the selected Triton kernel-cache artifact/configuration and autotuner record, whereas GDN calls bind eager source routes. It does not attest device-side completion, compiled GDN or underlying ATen/CUDA operators, driver/device binaries, a malicious runtime, or another model/runtime/hardware stack. The live-binding rerun strengthens ownership provenance only in one selected cell and six rows per phase; other factorial cells and coordinates retain the trace-relative binding boundary.

The 60-item Qasper/2WikiMQA quality cohort is the broadest direct answer-quality evidence in this paper, but it is a single archival validation run on one model/H20 stack. Its raw shards and outcome calculations replay, whereas the original mutable code directory and absence of pre-run full source/model-weight ledgers prevent a source-frozen regeneration claim. Its paired interval does not establish equivalence or noninferiority. It records neither TTFT nor Recall. The separate systems execution uses the indices-6–9 subset of this cohort and supplies TTFT/TPOT/throughput point estimates, not an independent replication, population uncertainty, or a speedup over full-prefix reuse; it also has no Recall measurement. The two panels are not pooled, and blank table cells preserve these gaps explicitly.

The auxiliary serving panel uses one request stream per engine and compares each cache-enabled framework only with its own disabled reference. It does not measure continuous batching, eviction, admission control, maximum concurrency, or best-tuned performance. CUDA graphs were disabled in both pinned runs; the reported values describe that fixed configuration only.

The HYPIC timing/quality comparison uses the authors' exact tracked commit on SGLang 0.5.14 in one fixed CUDA 12.9 / PyTorch 2.11 configuration, adapting Qwen3.5 execution from the paper's TP=2 to TP=1 so every arm uses one H20-3e per workload. Cache-hit authority comes from OpenAI response usage. Each of its 24 timing/quality cells has one formal measurement after a discarded, prefix-disjoint warmup. A separate receipt-instrumented run uses the same model/slice and fixed TP=1 serving configuration to produce 8 Prefix Cache and 8 HYPIC retained-state cells; each completed cell binds raw output plus pre-measurement and terminal ownership receipts, and all 16 pass an external hash-bound replay. The instrumentation adds target-entry byte-range receipts and a lifecycle-idempotency overlay; Store therefore does not come from the unmodified timing run. Neither cohort establishes scheduler throughput, QPS, capacity, the paper's TP=2 latency, or accuracy robustness beyond this eight-item slice.

The same-checkpoint related-work transfer is likewise operator-bounded: Hydragen uses one captured layer-3 shared-prefix attention family, and Palu uses one layer with eight calibration rows and one held-out row. Neither executes all model layers, a fused Palu kernel, LongBench, or a jointly optimized Hydragen/Palu-plus-CoMem serving path.

The page equations are deterministic identities, not learned laws; the primary factorial has one observation per rank–configuration cell and no stochastic replication. Most runtime tensors are represented by hashes and equality receipts rather than archived in full. The attention and GDN oracle rows are the exception and ship numerical sidecars, but cover only selected captured-input operator boundaries rather than an end-to-end model. The expanded numerical sweep covers two short inputs, all ten attention layers, and 12 of 30 GDN layers; it does not establish length/input robustness or full recurrent-layer coverage. Both the original and expanded GDN runs start after native q/k normalization and therefore do not validate that upstream boundary. Pointer-free normalized intervals plus raw-byte hashes make the primary captured storage evidence replayable. The preproducer census supplies the expected semantic slot set, and a second producer/observer execution reduces single-run capture risk, but both still trust correct slot-ID-to-live-tensor binding, IPC semantics, and paused capture; neither protects against a malicious producer. The selected-cell live binding directly checks the actual builder, serializer, and clone lineage, but shares the same honest runtime and does not generalize to unselected coordinates. The separate Falcon track compares against an official full-model reference only on its frozen Transformers/H20 naive path and supports no performance, memory, quality, or cross-runtime claim. The nine mutants are targeted positive controls; the target-suppression matrix adds a same-system comparison for those same designed faults, not blind or naturally occurring defects. Four complete with unchanged measured tokens and exact logits, one changes only full logits, and four stop earlier; N/O cells are not misses, while the injected M8 sentinel is not a production detector. This is not a general detection rate, a blind-fault result, or superiority over every conventional test suite. The scheduler extension's three frozen faults are expected-gate outcomes, not a detection rate or evidence of fault-set completeness. Broader systems claims would additionally require a native production serving scheduler, per-kernel overlap evidence, in-flight cancellation tests, optimized controls, alignment/length/input sweeps, and process-level capacity accounting.

The five designer–executor-separated controls were fixed before executor preparation, but remain deliberately constructed, fixed-stack mutations. Passing all five does not estimate unseen-fault recall, false-positive rate, or robustness to arbitrary runtime, model, schedule, or adversarial capture defects.

The retrospective case study concerns one organically encountered integration defect in our borrowed-state path. Only ranks 0–2 reproduce its archived input coordinates; ranks 3–7 are additional frozen inputs, and none is a statistically independent defect. The study does not estimate natural-defect prevalence or recall. Output and terminal-state differentials miss the alias, while a persistent-base content invariant catches it; FORKAUDIT contributes authenticated transition-time owner/layer/family localization rather than exclusive detection over all conventional testing.

J ARTIFACT AND CLAIM MAP

Artifact footprint. The source-complete primary package, derived summaries, raw ledgers, and expanded numerical sidecars are content-bound and locally replayable. The table below maps each manuscript claim to its machine-readable source and validation boundary; supporting measurements not used by a manuscript claim remain in the internal experiment registry.

Artifact aliases. Every package below is named by a stable neutral alias listed with its SHA-256 in the accompanying research package's `MANIFEST.json`. Aliases replace the internal directory names used during development; they carry no ordering, date, or platform information, and no path here resolves to an external service. A reviewer verifies each package by content hash rather than by path.

Table 30: Artifact and claim map. Relative paths are rooted at the accompanying research package. Completed rows authorize only their stated bounded claims. Metadata-redacted derivatives preserve numerical fields and state digests, while most full tensors still require the pinned environment for independent numerical recomputation.

Claim/evidence	Machine-readable source
Primary factorial replay package	`package/fa-01-primary-factorial/`
Replay and package manifest	`package/fa-01-primary-factorial/replay/run_replay.sh` and `package/fa-01-primary-factorial/MANIFEST.json`
Registered, derived, and raw evidence	`upstream/forkaudit-summary.json`, `derived/derived_summary_v2.json`, and `upstream/raw/` under the primary replay package
Executed source and storage tests	`executed_source/gpu/` and `replay/test_storage_witness.py` under the primary replay package
Identity, lifecycle, scheduler, and GDN extensions	`package/fa-08-terminal-closure/`, `evidence/lifecycle_transfer_reviewer_package/`, `evidence/round23_a2_scheduler_interleave/`, and `evidence/gdn_transition_oracle_preregistered_20260819d/`
Mac and H20 deployment controls	`package/qp-mac-motivation/`, `package/qp-08-timing/`, and their generators under `scripts/`
Q-CoMem 60-item Store–F1 validation	`package/qp-60-store-f1/`; exact raw-outcome replay: `replay/run_replay.sh`; numerical summary and claim boundary: `validation_report.json` and `claim_boundary.json`; registry entry `E-QCOMEM-60-VALIDATION-A`
HYPIC retained-state receipts	`evidence/experiment_registry.json`, entry `E-HYPIC-STORE-EXTERNAL-ACCEPTANCE`

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Table 30 continued.

Claim/evidence	Machine-readable source
Target-gate-suppression preregistration and raw ledger	`package/fa-02-gate-suppression/formal_run_20260824a/preregistration/detector-matrix-v2-plan.json` `package/fa-02-gate-suppression/formal_run_20260824a/receipts/raw-artifacts.sha256`
Target-gate-suppression summary and replay receipt	`package/fa-02-gate-suppression/formal_run_20260824a/detector-matrix-v2-summary.postexec-corrected.json` `package/fa-02-gate-suppression/formal_run_20260824a/postexecution-correction-replay-receipt.json`; disclosed field-source amendment: `package/fa-02-gate-suppression/postexecution-rr2-run-binding-correction.json`
vLLM, SGLang, HYPIC timing/quality, and Marconi controls	`package/rw-serving-controls/`; exact package members and one-command replays reside under this directory
Hydragen/Palu same-checkpoint transfers	`package/rw-operator-transfer/` and its `executed_source/` directory
Published context and table generator	`literature/reported_system_context.json` and `scripts/generate_related_work_context_table.py`
Out-of-process GDN observer	`package/fa-06-slot-census/formal_h20/result/preregistration/preregistration.json` `package/fa-06-slot-census/formal_h20/result/raw/out-of-process-gdn-capture.json` `package/fa-06-slot-census/formal_h20/result/replay/out-of-process-gdn-replay.json` `package/fa-06-slot-census/formal_h20/result/receipts/terminal-files.sha256`
Independent preproducer slot census	`evidence/experiment_registry.json`, entry `E-SLOT-CENSUS-A`; execution package, formal archive, preexecution receipt, aggregate, controls, and terminal ledger are path- and SHA-bound there
Dual-producer repeat	`evidence/experiment_registry.json`, entry `E-DUAL-PRODUCER-REPEAT-A`; execution package, formal archive/sidecar, summary digest, and exact replay boundary are bound there
Falcon-H1 bounded transfer	`evidence/experiment_registry.json`, entry `E-FALCON-H1-TRANSFER-V2-A`; formal archive/sidecar, count clarification, validation, aggregate, and terminal ledger are path- and SHA-bound there
Primary compiled dispatch	`evidence/experiment_registry.json`, entry `E-COMPILED-DISPATCH-A`; the immutable source package retains its pre-run HOLD metadata, while the separately hash-bound post-run aggregate and mirror authorize the bounded target-5 result
Selected-cell live binding	`evidence/experiment_registry.json`, entry `E-LIVE-BINDING-A`, and `package/fa-09-live-binding-audit-mirror/`; the immutable package retains pre-run metadata, while the terminal result and hash-identical post-run audit authorize only the bounded selected-cell claim

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Table 30 continued.

Claim/evidence	Machine-readable source
Designer–executor-separated faults	`package/fa-05-separated-faults/author_freeze/designer_input/prior-submission.pdf` `package/fa-05-separated-faults/author_freeze/PROTOCOL.md` `package/fa-05-separated-faults/executor_attempt_b/AMENDMENT.md` `package/fa-05-separated-faults/formal_h20/RESULT_VERIFICATION.json` `package/fa-05-separated-faults/formal_h20/r33-fresh-faults-20260825b/rank-run-*/rank-*/mutant-case.json` `package/fa-05-separated-faults/formal_h20/r33-fresh-faults-20260825b/summary.json` `package/fa-05-separated-faults/formal_h20/r33-fresh-faults-20260825b/terminal-files.sha256`
Historical alias regression and repair	`package/fa-07-alias-regression/design_decision.md` `package/fa-07-alias-regression/preregistration.json` `package/fa-07-alias-regression/preexecution-freeze-receipt.json` `package/fa-07-alias-regression/preexecution-resource-amendment.json` `package/fa-07-alias-regression/formal_h20/RESULT_VERIFICATION.json` `package/fa-07-alias-regression/formal_h20/r35-historical-alias-20260826a/aggregate.json` `package/fa-07-alias-regression/formal_h20/r35-historical-alias-20260826a/replay/`
Bounded two-stream lifecycle	`package/fa-03-two-stream/formal_run_20260825b/raw/formal-result.json` `package/fa-03-two-stream/formal_run_20260825b/replay/independent-replay.json` `package/fa-03-two-stream/formal_run_20260825b/receipts/raw-and-replay.sha256`
Expanded captured-boundary numerical sweep	`package/fa-04-numerical-sweep/`; 272 local arrays under `raw/sidecars/`; canonical result `oracle-result.json`; local replay source `r30_expanded_oracle_reference.py`; validation `validation_report.json`; terminal binding `terminal-products.sha256`
Local replay and storage accounting	`evidence/experiment_registry.json`, entry `E-LOCAL-REPLAY-STORAGE-COST-V1-A`; protocol, Attempt-B amendment, aggregate, independent audit, and terminal ledger reside under `evidence/r40_ci_cost_accounting_v1/`
Manuscript claim index	`evidence/claim_evidence_map.tsv`